

Supplementary Material for “Innovation-Normalized Detection in Compound-Gaussian Clutter: Exact Laws, Conformal Thresholds and Certified Integration under Pulsed Interference”

M. U. Raza, S. Ahmed, and A. Ahmed

All numbers below are generated from the saved outcomes in the project repository (`research/r7c`). Seven protocols were frozen and hash-recorded before their outcomes were computed: the main IPIX study (`PROTOCOL.md`), single-pulse detection (`detection/PROTOCOL_DETECTION.md`), onset (`detection/PROTOCOL_ONSET.md`), baselines (`baselines/PROTOCOL_BASELINES.md`), the second radar (`jku/PROTOCOL_JKU.md`), classical detectors, gradual emergence and ACI contamination (`r10/PROTOCOL_R10.md`), order-statistic normalization with real interference (`r11/PROTOCOL_R11.md`), and low false-alarm calibration, ANMF baselines, the real IPIX target and certified integration under pulsed interference (`r12/PROTOCOL_R12.md`). A further protocol (`sdrdsp/PROTOCOL_SDRDSP.md`) was frozen before the SDRDSP data were obtained.

1 Data exposure

Table 1 states which data each stage used, so that the strength of each validation claim can be judged. Only the hold-out series and the second radar were not used while the methods were being developed.

Table 1: Data exposure by stage. “Seen” means that outcomes on these data were computed while methods, orders or comparisons could still change.

Data	Used for	Status of the reported results
IPIX, 14 sessions, rotation 1	Method development (R1–R7): method list, model orders, comparisons	Seen; not reported
IPIX, same sessions, rotations 2–3	Main study, detection, onset, baselines, classical detectors, order-statistic scores (R7–R11)	Frozen-protocol evaluation, computed once per protocol; the recordings were seen in rotation 1, so this is not an untouched confirmation
IPIX, day transfer	Transfer and ACI analyses	Frozen-protocol evaluation on the same recordings
IPIX #269, #287	Hold-out	Hashed and unopened until the main analysis was complete; evaluated once (small test sets)
JKU 77 GHz, reference run	Second-radar study	Not used in development; frozen protocol; the general-form gain curve is post hoc
JKU 77 GHz, interference runs A and C	Real-interference test (R11); certified integration (R12)	R11: only chirp-hit statistics inspected before its protocol. R12: R11 outcomes on these runs had been seen
IPIX, primary target cells	Real-target study (R12)	Excluded from every earlier study; frozen protocol, evaluated once
SDRDSP (pending)	Untouched sea-clutter confirmation	Protocol frozen and hashed before the data were obtained

2 Hold-out series

Two further McMaster series (#269 and #287, the widely used “high” and “low” sea-state files, one provider-preprocessed VV cell each, from the same 1993 campaign) were hashed and left unopened until the main analysis was complete, and then evaluated once (Table 2). They confirm the direction of the main result, with larger gains. They also show a case in which the classical Gaussian threshold over-covers while conformal IN-ARCP stays near nominal. Their test sets are

small (about 165 episodes per unit), so the hold-out is a directional check. Mondrian calibration returns infinite regions at $1 - \alpha = 0.99$ there, because each bin receives fewer calibration episodes than the rank requires.

Table 2: Hold-out series #269 and #287 ($1 - \alpha = 0.90$; six units per column).

Method	$m = 8$		$m = 16$	
	Radius ratio	Coverage (min)	Radius ratio	Coverage (min)
Noise-aware AR(4) (proposed)	0.454	0.890 (0.816)	0.498	0.893 (0.847)
Noise-aware AR(2) (proposed)	0.541	0.887 (0.828)	0.533	0.895 (0.859)
AR(4), innovation-normalized	0.546	0.908 (0.847)	0.515	0.906 (0.883)
AR(2), innovation-normalized	0.582	0.905 (0.840)	0.572	0.906 (0.871)
Noise-aware AR(1)	0.921	0.898 (0.840)	0.921	0.903 (0.859)
IN-ARCP, AR(1)	1.000	0.906 (0.853)	1.000	0.910 (0.859)
Gaussian plug-in of IN-ARCP	1.178	0.949 (0.896)	1.104	0.945 (0.895)
Mondrian IN-ARCP	1.000	0.895 (0.804)	0.981	0.896 (0.810)
Learned-scale CP	0.629	0.893 (0.804)	0.662	0.919 (0.791)
Invariant MLP CP	1.809	0.884 (0.736)	2.158	0.903 (0.761)
Raw-RMS normalized CP	1.321	0.880 (0.706)	1.243	0.887 (0.706)
Unnormalized CP	1.142	0.868 (0.607)	1.161	0.859 (0.589)
Gaussian plug-in of noise-aware AR(4)	0.411	0.866 (0.816)	0.472	0.856 (0.755)

3 Synthetic verification

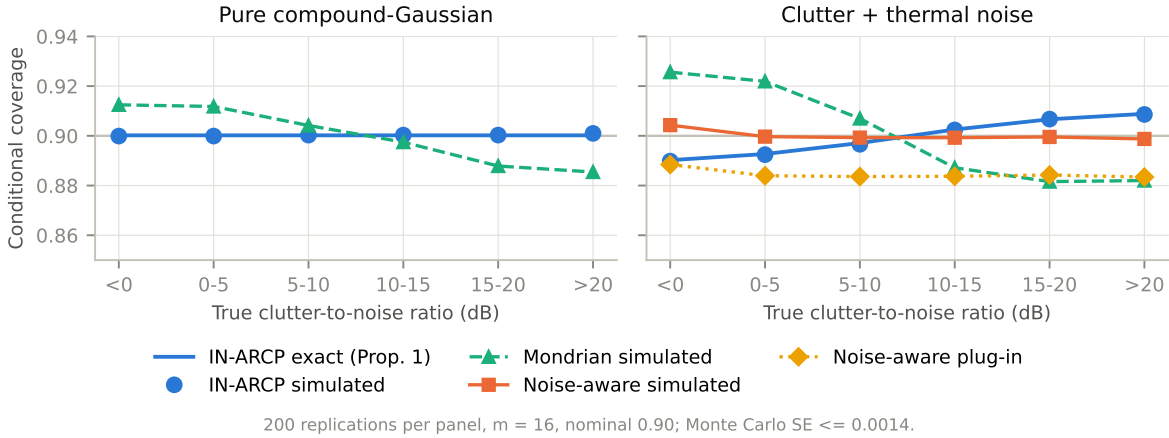


Figure 1: Synthetic verification: coverage conditional on the true clutter-to-noise ratio. The line is the exact law of Proposition 1; markers are simulation.

4 Proof of Proposition 2 and the latent-texture Mondrian result

Equation, proposition and theorem numbers without the prefix S refer to the paper.

Proposition 2. If $e \sim \mathcal{CN}(0, v)$ and a Swerling-1 target of power SP is added, a threshold test on $|e|^2/v$ at false-alarm probability α detects with probability $\alpha^{1/(1+SP/v)}$, while the power-only test on $|Y|^2/P$ detects with $\alpha^{1/(1+S)}$; equal P_d therefore requires SCRs in the ratio $G = P/v$. For the centre rH_m under (1) with AR(1) speckle, $v = \nu\{c(1 + |r|^2 - 2\Re\bar{r}\rho) + 1 + |r|^2\}$, which gives G_{IN} . For the optimal predictor from the infinite past, v is the one-step prediction-error variance of the AR(1)-plus-white-noise process, $\exp\{\int \log f\}$ by the Kolmogorov-Szegő formula (Brockwell and Davis, *Time Series: Theory and Methods*, 1991). With $f(\omega) = \nu\{c(1 - |\rho|^2) + |1 - \rho e^{-j\omega}|^2\}/|1 - \rho e^{-j\omega}|^2$, the denominator contributes zero, and the numerator factorizes as $\nu K|1 - \beta' e^{-j\omega}|^2$ with $K(1 + |\beta'|^2) = \beta$, $K\beta' = \rho$ and $|\beta'| < 1$, which gives $v = \nu K$ and G_{opt} . Because the optimal predictor's error variance is at most that of any linear predictor, including 0 and rH_m , $G_{\text{opt}} \geq \max\{1, G_{\text{IN}}\}$.

Latent-texture Mondrian calibration. Mondrian (group-conditional) conformal prediction calibrates separately within bins of an observed statistic and is valid conditional on the observed bin; Dewolf et al. report that estimated taxonomies mix up the true classes. For a latent texture the consequence can be stated exactly.

Proposition S1. *Let $\nu = 0$ and $X = \sigma Z$, with $\sigma > 0$ distributed as G , non-degenerate and independent of $Z \sim \mathcal{CN}(0, R)$. Let $S(Z)$ be a scale-invariant score with a continuous distribution function F_S that is strictly increasing on its support, let $T(H) > 0$ be continuous and homogeneous of degree one, and calibrate within fixed bins $B_b = [e_{b-1}, e_b)$, $0 = e_0 < \dots < e_K = \infty$, $K \geq 2$, of $T(H)$ with population quantiles q_b . Let $\mathcal{C}(\sigma)$ be the coverage conditional on the texture. (i) If $S(Z)$ is independent of $T(Z_H)$, then $\mathcal{C}(\sigma) \equiv 1 - \alpha$. (ii) If $S(Z)$ given $T(Z_H) = t$ is strictly stochastically decreasing in t , then $\lim_{\sigma \rightarrow \infty} \mathcal{C}(\sigma) = F_S(q_K) < 1 - \alpha < F_S(q_1) = \lim_{\sigma \rightarrow 0} \mathcal{C}(\sigma)$, whereas the unbinned threshold gives a constant $\mathcal{C}(\sigma)$. Condition (ii) holds exactly for IN-ARCP at the true coefficient with $T = s_\rho$.*

Proof. Write $t = T(Z_H)$, so that the observed statistic is $T(X_H) = \sigma t$ and $S(X) = S(Z)$. For bin b let $w_b(t) = \mathbb{P}\{\sigma t \in B_b\}$. Since σ is independent of Z , for every x

$$\mathbb{P}\{S \leq x, \sigma t \in B_b\} = \mathbb{E}[\varphi_x(t) w_b(t)], \quad (\text{S1})$$

where $\varphi_x(t) = \mathbb{P}\{S \leq x \mid t\}$. (i) If S is independent of t , then $\varphi_x \equiv F_S(x)$, every bin-conditional law of S equals F_S , all q_b equal $F_S^{-1}(1 - \alpha)$ and $\mathcal{C}(\sigma) = 1 - \alpha$.

(ii) Let $x_0 = F_S^{-1}(1 - \alpha)$. For the top bin, $w_K(t) = \mathbb{P}\{\sigma \geq e_{K-1}/t\}$ is nondecreasing in t . Because T is continuous, positive and homogeneous and Z_H has full support, t has full support on $(0, \infty)$; as e_{K-1}/t then ranges over $(0, \infty)$ and G is non-degenerate, $w_K(t)$ is not almost surely constant. By assumption φ_{x_0} is strictly increasing. Let t' be an independent copy of t , $\Delta_\varphi = \varphi_{x_0}(t) - \varphi_{x_0}(t')$ and $\Delta_w = w_K(t) - w_K(t')$. Then

$$\text{Cov}\{\varphi_{x_0}(t), w_K(t)\} = \frac{1}{2} \mathbb{E}[\Delta_\varphi \Delta_w] > 0,$$

because $\Delta_\varphi \Delta_w \geq 0$, with strict inequality on the event $\{w_K(t) \neq w_K(t')\}$, which has positive probability. By (S1), $\mathbb{P}\{S \leq x_0 \mid \sigma t \in B_K\} = \mathbb{E}[\varphi_{x_0} w_K] / \mathbb{E}[w_K] > \mathbb{E}[\varphi_{x_0}] = 1 - \alpha$. The bin-conditional distribution function of S is continuous, so there is $x_1 < x_0$ at which it still exceeds $1 - \alpha$; hence $q_K \leq x_1 < x_0$ and, F_S being strictly increasing, $F_S(q_K) < 1 - \alpha$. For the bottom bin, $w_1(t) = \mathbb{P}\{\sigma < e_1/t\}$ is nonincreasing, the covariance is negative, and the same steps give $q_1 > x_0$ and $F_S(q_1) > 1 - \alpha$.

The coverage at texture σ is $\mathcal{C}(\sigma) = \sum_b \mathbb{P}\{S \leq q_b, \sigma t \in B_b\}$. As $\sigma \rightarrow \infty$, $\sigma t \rightarrow \infty$ almost surely, so $\mathbf{1}\{\sigma t \in B_K\} \rightarrow 1$ and, by dominated convergence, $\mathcal{C}(\sigma) \rightarrow F_S(q_K)$; as $\sigma \rightarrow 0$, $\mathcal{C}(\sigma) \rightarrow F_S(q_1)$. An unbinned threshold q gives $\mathcal{C}(\sigma) = F_S(q)$ for every σ .

Finally, for IN-ARCP at $r = \rho$ and $T = s_\rho$, $S = |E|/t$ with the innovation E independent of Z_H , so $\varphi_x(t) = \mathbb{P}\{|E| \leq xt\}$ is strictly increasing in t , which is condition (ii). $\blacksquare$

5 Zero-noise limit of the noise-aware score

Proposition S2. *Let R be $AR(1)$ with coefficient ρ and fix a nonzero history h . As $\nu \downarrow 0$, $\nu \hat{c}(h) \rightarrow h^H R_H^{-1} h/m$, $\mu(h) \rightarrow \rho h_m$ and $s(h) \rightarrow s_\rho(h)$, so the NA score converges to the IN-ARCP score at $r = \rho$.*

Proof sketch. With $\gamma = \nu c$ the history log-likelihood converges locally uniformly on $\gamma > 0$ to $-m \log \gamma - \gamma^{-1} h^H R_H^{-1} h$, whose unique maximizer is $h^H R_H^{-1} h/m$; coercivity at 0 and ∞ gives convergence of the maximizers. The Markov property gives $\mu \rightarrow \rho h_m$, and $(1 - |\rho|^2) R_H^{-1} = W_\rho^H W_\rho$, with W_ρ the bidiagonal whitening matrix, identifies the limiting predictive variance with $s_\rho(h)^2$.

6 Texture-conditional coverage and disc widths on IPIX

7 Day transfer and texture-proxy sensitivity

Texture-proxy sensitivity (pre-specified windows; mean maximum quintile deviation at $1 - \alpha = 0.90$, $m = 16$): with ± 512 pulses IN-ARCP 0.035, NA-AR(4) 0.043, unnormalized 0.181, neural centre 0.103; with ± 2048 pulses IN-ARCP 0.033, unnormalized 0.141. The ranking is the same as with ± 1024 .

8 Additional figures for the IPIX study

9 Detection study: all levels and methods

Pooled detection probability by SCR, measured false-alarm rates, exact-law predictions and SCR gains with 95% day-cluster intervals (output of `detection/analyze_detection.py`):

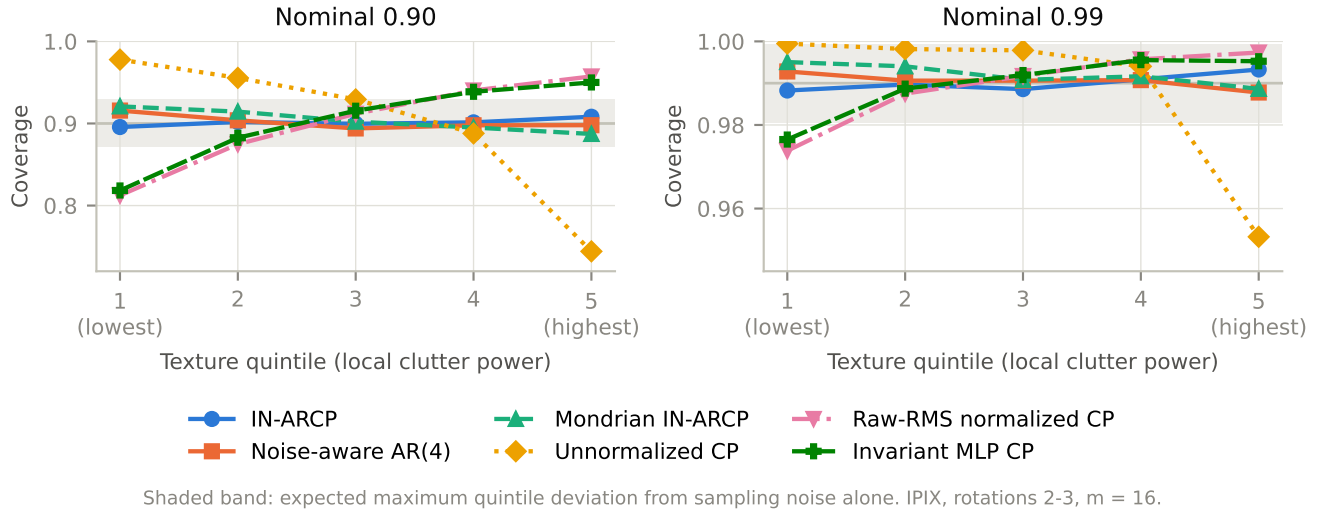


Figure 2: Coverage by texture quintile on IPIX (rotations 2–3, $m = 16$). Texture is proxied by the local clutter power (± 1024 pulses, excluding the episode).

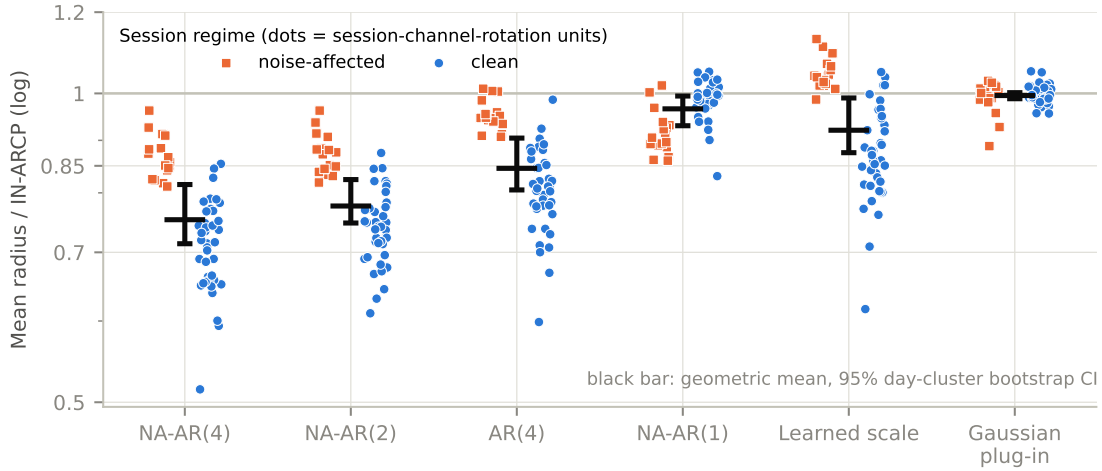


Figure 3: Per-unit radius relative to IN-ARCP at $1 - \alpha = 0.90$, $m = 16$, by session regime (noise-affected: fitted noise power above 1% of mean power).

```
units: 56; days: 6; test episodes: 74980

=== 1-alpha = 0.9 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
CALoc 0.1030 0.166 0.219 0.307 0.433 0.579 0.711 0.814 0.888 0.934 0.963 0.979 0.988 0.993
CA16 0.1002 0.242 0.312 0.406 0.516 0.631 0.736 0.823 0.887 0.931 0.959 0.977 0.987 0.992
U 0.1011 0.446 0.549 0.651 0.745 0.823 0.883 0.926 0.955 0.974 0.985 0.992 0.995 0.997
RMS 0.1007 0.442 0.541 0.642 0.734 0.813 0.877 0.922 0.953 0.972 0.984 0.991 0.995 0.997
MLP 0.0990 0.447 0.548 0.650 0.743 0.823 0.884 0.927 0.956 0.974 0.985 0.992 0.995 0.997
LS 0.0995 0.534 0.624 0.710 0.784 0.847 0.898 0.935 0.960 0.977 0.987 0.993 0.996 0.997
MDN 0.0961 0.540 0.634 0.720 0.794 0.858 0.907 0.941 0.965 0.979 0.988 0.993 0.996 0.998
G1 0.1004 0.542 0.635 0.722 0.795 0.859 0.908 0.942 0.965 0.980 0.989 0.993 0.996 0.998
IN1 0.0987 0.540 0.633 0.720 0.793 0.857 0.906 0.941 0.964 0.979 0.988 0.993 0.996 0.998
IN2 0.0991 0.553 0.643 0.728 0.801 0.862 0.909 0.943 0.966 0.980 0.989 0.993 0.996 0.998
IN4 0.0985 0.574 0.661 0.741 0.811 0.869 0.914 0.946 0.967 0.981 0.989 0.994 0.996 0.998
NA1 0.0994 0.553 0.648 0.735 0.810 0.872 0.917 0.948 0.969 0.982 0.990 0.994 0.997 0.998
NA2 0.0987 0.609 0.692 0.767 0.834 0.887 0.926 0.954 0.972 0.984 0.991 0.995 0.997 0.998
NA4 0.0982 0.614 0.696 0.770 0.836 0.888 0.927 0.954 0.973 0.984 0.991 0.995 0.997 0.998
NA4G 0.1150 0.630 0.710 0.782 0.845 0.895 0.932 0.958 0.975 0.986 0.992 0.995 0.997 0.998
IN1 exact-law prediction 0.507 0.614 0.710 0.791 0.856 0.906 0.942 0.965 0.979 0.988 0.993 0.996 0.998
exact-law MAE, IN1 Pd: per unit x SCR 0.0140; pooled curve 0.0053
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR -3.0 dB | vs CA16 5.06 [ 3.25, 6.46] | vs CALoc 6.64 [ 4.50, 8.41]
IN4 SCR -3.1 dB | vs CA16 5.19 [ 3.41, 6.53] | vs CALoc 6.76 [ 4.69, 8.46]
NA4 SCR -3.4 dB | vs CA16 5.41 [ 3.85, 6.61] | vs CALoc 6.99 [ 5.10, 8.52]
G1 SCR -3.1 dB | vs CA16 5.10 [ 3.35, 6.44] | vs CALoc 6.67 [ 4.65, 8.39]
U SCR -2.6 dB | vs CA16 4.61 [ 2.79, 6.04] | vs CALoc 6.19 [ 4.15, 8.03]
MLP SCR -2.6 dB | vs CA16 4.66 [ 2.72, 6.11] | vs CALoc 6.23 [ 3.87, 8.10]
LS SCR -2.8 dB | vs CA16 4.83 [ 2.83, 6.34] | vs CALoc 6.40 [ 4.06, 8.30]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 5.2 dB | vs CA16 7.98 [ 5.03, 10.43] | vs CALoc 7.93 [ 5.02, 10.37]
IN4 SCR 4.3 dB | vs CA16 8.86 [ 5.48, 11.49] | vs CALoc 8.80 [ 5.45, 11.44]
NA4 SCR 3.1 dB | vs CA16 10.05 [ 6.52, 12.77] | vs CALoc 9.99 [ 6.51, 12.72]
G1 SCR 5.1 dB | vs CA16 8.02 [ 5.09, 10.42] | vs CALoc 7.97 [ 5.11, 10.38]
```

Table 3: IPIX width and coverage ($m = 16$; rotations 2–3; 56 units). Radius ratio: geometric mean over units of mean disc radius relative to IN-ARCP on the same test episodes, with 95% day-cluster bootstrap interval. Max. dev.: mean over units of the largest absolute texture-quintile coverage deviation from nominal. At equal model order, NA-AR(4) has 0.891 [0.872, 0.905] times the radius of the innovation-normalized AR(4) disc.

Method	$1 - \alpha = 0.90$			$1 - \alpha = 0.99$		
	Radius ratio [95% CI]	Coverage	Max. dev.	Radius ratio [95% CI]	Coverage	Max. dev.
Noise-aware AR(4) (proposed)	0.753 [0.714, 0.815]	0.902	0.043	0.743 [0.706, 0.798]	0.991	0.011
Noise-aware AR(2) (proposed)	0.777 [0.748, 0.824]	0.901	0.044	0.769 [0.745, 0.806]	0.991	0.011
AR(4), innovation-normalized	0.845 [0.805, 0.905]	0.901	0.032	0.861 [0.822, 0.918]	0.990	0.011
AR(2), innovation-normalized	0.895 [0.855, 0.953]	0.901	0.034	0.898 [0.857, 0.953]	0.990	0.010
Noise-aware AR(1)	0.966 [0.930, 0.994]	0.901	0.042	0.952 [0.907, 0.989]	0.990	0.011
IN-ARCP, AR(1)	1	0.901	0.036	1	0.990	0.011
Gaussian plug-in of IN-ARCP	0.995 [0.988, 1.002]	0.900	0.034	0.993 [0.979, 1.005]	0.990	0.011
Mondrian IN-ARCP	0.980 [0.975, 0.984]	0.904	0.043	0.998 [0.988, 1.006]	0.992	0.011
Learned-scale CP	0.921 [0.875, 0.990]	0.900	0.039	1.072 [1.019, 1.147]	0.991	0.012
Invariant MLP CP	1.337 [1.288, 1.439]	0.901	0.096	1.585 [1.489, 1.780]	0.990	0.019
Raw-RMS normalized CP	1.368 [1.320, 1.449]	0.899	0.103	1.609 [1.533, 1.782]	0.989	0.021
Unnormalized CP	1.048 [1.010, 1.085]	0.899	0.165	1.475 [1.261, 1.720]	0.989	0.039
Gaussian plug-in of noise-aware AR(4)	0.726 [0.687, 0.787]	0.885	0.049	0.698 [0.659, 0.758]	0.985	0.016
Sampling-noise reference for max. dev.			0.029			0.009

Table 4: Day transfer ($m = 16$, $1 - \alpha = 0.90$): fit and calibrate on other days, test on each of six held-out days.

Method	Mean coverage	Worst held-out day	Radius ratio
Noise-aware AR(4) (proposed)	0.898	0.863	0.936
Noise-aware AR(2) (proposed)	0.898	0.883	0.956
AR(4), innovation-normalized	0.901	0.894	0.993
AR(2), innovation-normalized	0.899	0.891	1.024
Noise-aware AR(1)	0.900	0.882	1.037
IN-ARCP, AR(1)	0.903	0.882	1.000
Gaussian plug-in of IN-ARCP	0.906	0.893	1.006
Mondrian IN-ARCP	0.907	0.881	0.984
Learned-scale CP	0.898	0.774	0.967
Invariant MLP CP	0.880	0.622	1.200
Raw-RMS normalized CP	0.906	0.681	1.466
Unnormalized CP	0.909	0.716	1.470
Gaussian plug-in of noise-aware AR(4)	0.876	0.842	0.893

```

U   SCR   7.1 dB | vs CA16 6.09 [ 3.33, 8.23] | vs Caloc 6.04 [ 3.30, 8.21]
MLP  SCR   7.0 dB | vs CA16 6.12 [ 3.70, 7.88] | vs Caloc 6.07 [ 3.70, 7.81]
LS   SCR   5.4 dB | vs CA16 7.78 [ 4.32,10.40] | vs Caloc 7.72 [ 4.35,10.37]

=== 1-alpha = 0.99 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===
method Pfa_meas -5.0 -2.5 0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0 22.5 25.0
Caloc 0.0110 0.022 0.034 0.062 0.125 0.237 0.394 0.566 0.714 0.822 0.893 0.939 0.966 0.980
CA16 0.0103 0.081 0.129 0.203 0.306 0.434 0.568 0.695 0.796 0.872 0.922 0.954 0.974 0.985
U 0.0115 0.131 0.208 0.308 0.428 0.558 0.680 0.782 0.860 0.913 0.948 0.970 0.983 0.990
RMS 0.0108 0.176 0.271 0.383 0.503 0.625 0.733 0.822 0.887 0.931 0.960 0.977 0.987 0.993
MLP 0.0104 0.176 0.272 0.388 0.510 0.632 0.742 0.831 0.893 0.935 0.962 0.978 0.988 0.993
LS 0.0095 0.288 0.391 0.495 0.597 0.694 0.776 0.846 0.900 0.938 0.963 0.979 0.988 0.993
MDN 0.0080 0.326 0.434 0.546 0.651 0.743 0.822 0.883 0.927 0.956 0.974 0.985 0.992 0.995
G1 0.0101 0.337 0.448 0.558 0.661 0.751 0.828 0.887 0.929 0.957 0.975 0.986 0.992 0.995
IN1 0.0099 0.335 0.445 0.554 0.657 0.747 0.825 0.885 0.927 0.956 0.975 0.986 0.992 0.995
IN2 0.0102 0.357 0.464 0.569 0.670 0.758 0.832 0.889 0.931 0.959 0.976 0.986 0.992 0.996
IN4 0.0098 0.377 0.482 0.586 0.681 0.765 0.836 0.892 0.932 0.960 0.977 0.987 0.992 0.996
NA1 0.0099 0.354 0.466 0.577 0.681 0.773 0.848 0.903 0.940 0.965 0.980 0.989 0.994 0.996
NA2 0.0095 0.423 0.531 0.632 0.723 0.802 0.868 0.915 0.948 0.969 0.982 0.990 0.994 0.997
NA4 0.0095 0.433 0.539 0.637 0.727 0.805 0.869 0.916 0.948 0.969 0.982 0.990 0.994 0.997
NA4G 0.0151 0.465 0.566 0.659 0.745 0.819 0.878 0.921 0.952 0.971 0.984 0.991 0.995 0.997
IN1 exact-law prediction 0.281 0.406 0.531 0.645 0.743 0.824 0.885 0.929 0.957 0.975 0.986 0.992 0.995
exact-law MAE, IN1 Pd: per unit x SCR 0.0210; pooled curve 0.0105
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR -0.9 dB | vs CA16 6.96 [ 4.29, 9.04] | vs Caloc 9.81 [ 6.41,12.61]
IN4 SCR -1.3 dB | vs CA16 7.37 [ 4.54, 9.52] | vs Caloc 10.22 [ 6.67,13.26]
NA4 SCR -1.9 dB | vs CA16 8.03 [ 5.45,10.02] | vs Caloc 10.88 [ 7.56,13.66]
G1 SCR -1.0 dB | vs CA16 7.05 [ 4.40, 9.11] | vs Caloc 9.90 [ 6.48,12.71]
U SCR 3.8 dB | vs CA16 2.30 [ 0.78, 3.44] | vs Caloc 5.15 [ 3.34, 6.61]
MLP SCR 2.2 dB | vs CA16 3.88 [ 1.89, 5.34] | vs Caloc 6.73 [ 3.89, 8.70]
LS SCR 0.4 dB | vs CA16 5.72 [ 2.75, 7.96] | vs Caloc 8.57 [ 4.89,11.49]
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --
IN1 SCR 8.7 dB | vs CA16 7.62 [ 4.55,10.01] | vs Caloc 8.97 [ 5.58,11.67]
IN4 SCR 8.0 dB | vs CA16 8.35 [ 4.95,10.96] | vs Caloc 9.70 [ 5.90,12.64]
NA4 SCR 6.5 dB | vs CA16 9.81 [ 6.27,12.45] | vs Caloc 11.16 [ 7.28,14.14]
G1 SCR 8.7 dB | vs CA16 7.67 [ 4.71,10.02] | vs Caloc 9.02 [ 5.70,11.73]
U SCR 13.5 dB | vs CA16 2.80 [ 1.01, 4.34] | vs Caloc 4.15 [ 2.23, 5.73]
MLP SCR 12.0 dB | vs CA16 4.36 [ 2.30, 5.85] | vs Caloc 5.71 [ 3.23, 7.35]
LS SCR 10.2 dB | vs CA16 6.14 [ 2.75, 8.69] | vs Caloc 7.49 [ 3.78,10.46]

=== 1-alpha = 0.999 (Swerling 1; Pd pooled mean over units; SCR in dB rel. local clutter+noise power) ===

```

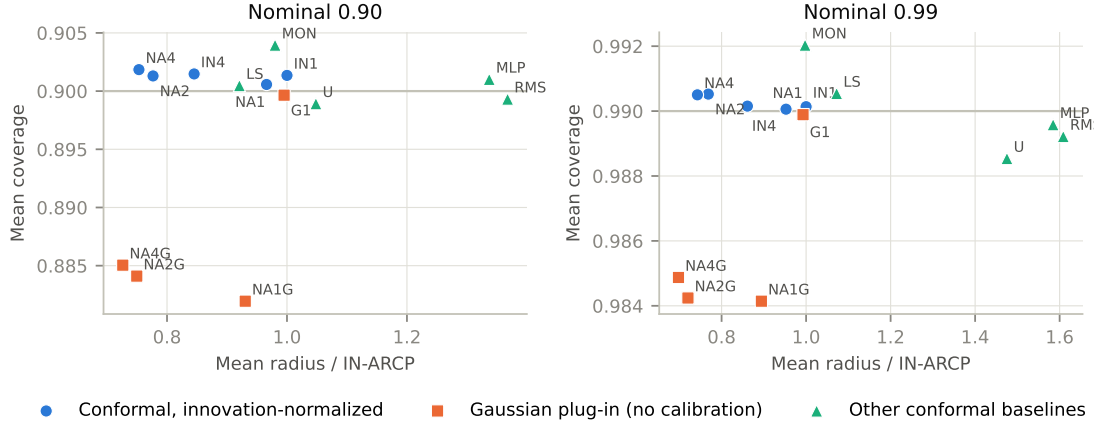


Figure 4: Mean coverage against mean radius (confirmatory, $m = 16$). Gaussian plug-ins of the richer noise-aware models are shorter but under-cover; conformal calibration restores nominal coverage. Labels: NA_p and IN_p , noise-aware and innovation-normalized $AR(p)$ ($IN1$ is IN -ARCP); $G1$ and NA_pG , Gaussian plug-ins; MON , Mondrian; LS , learned scale; U , unnormalized; RMS , raw-RMS; MLP , neural centre.

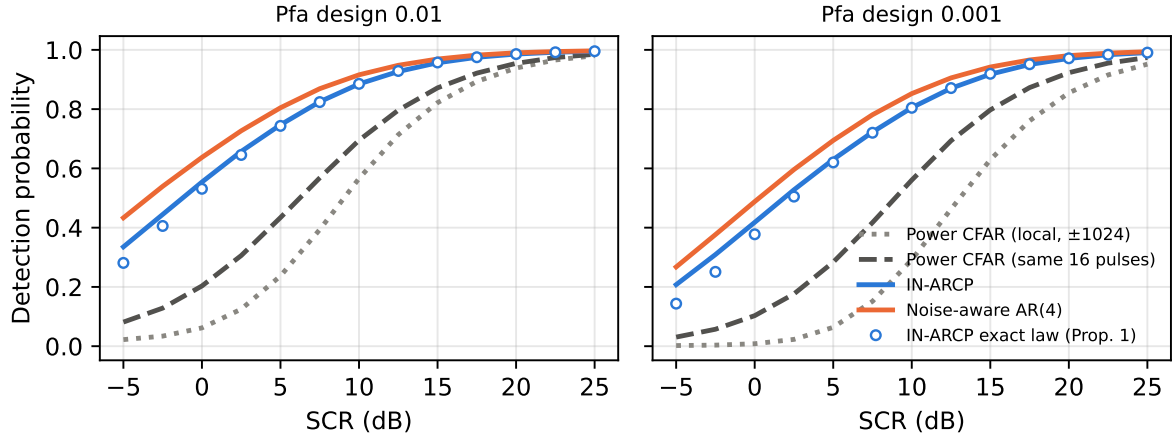


Figure 5: Detection of injected Swerling-1 targets on IPIX at design false-alarm probabilities 0.01 and 0.001.

method	Pfa_meas	-5.0	-2.5	0.0	2.5	5.0	7.5	10.0	12.5	15.0	17.5	20.0	22.5	25.0
Caloc	0.0008	0.002	0.004	0.008	0.023	0.064	0.151	0.292	0.463	0.629	0.761	0.855	0.915	0.952
CA16	0.0007	0.031	0.057	0.103	0.176	0.284	0.417	0.561	0.693	0.797	0.873	0.923	0.955	0.974
U	0.0014	0.023	0.048	0.092	0.159	0.251	0.362	0.485	0.606	0.710	0.796	0.863	0.911	0.944
RMS	0.0007	0.057	0.108	0.184	0.287	0.411	0.539	0.665	0.772	0.853	0.910	0.946	0.969	0.982
MLP	0.0009	0.057	0.110	0.190	0.297	0.426	0.559	0.683	0.788	0.865	0.918	0.952	0.972	0.984
LS	0.0010	0.124	0.198	0.290	0.390	0.494	0.596	0.693	0.778	0.849	0.902	0.940	0.964	0.979
MON	0.0000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
G1	0.0013	0.226	0.333	0.446	0.560	0.665	0.756	0.834	0.893	0.934	0.961	0.977	0.987	0.993
IN1	0.0011	0.208	0.309	0.418	0.528	0.630	0.723	0.805	0.870	0.918	0.951	0.971	0.984	0.991
IN2	0.0009	0.209	0.309	0.418	0.526	0.629	0.722	0.803	0.868	0.916	0.950	0.970	0.983	0.990
IN4	0.0008	0.230	0.332	0.440	0.543	0.641	0.729	0.804	0.866	0.913	0.947	0.969	0.982	0.990
NA1	0.0012	0.219	0.325	0.440	0.554	0.661	0.758	0.837	0.897	0.937	0.963	0.979	0.988	0.993
NA2	0.0009	0.272	0.380	0.491	0.598	0.695	0.783	0.853	0.906	0.943	0.966	0.981	0.989	0.994
NA4	0.0007	0.267	0.377	0.488	0.596	0.694	0.781	0.852	0.905	0.942	0.966	0.981	0.989	0.994
NA4G	0.0028	0.370	0.477	0.581	0.674	0.760	0.833	0.890	0.931	0.958	0.976	0.986	0.992	0.996
IN1 exact-law prediction 0.144 0.251 0.378 0.505 0.620 0.720 0.804 0.871 0.919 0.951 0.972 0.984 0.991														
exact-law MAE, IN1 Pd: per unit x SCR 0.0266; pooled curve 0.0155														
-- SCR gain (dB) needed for Pd=0.5; positive = less SCR than reference; 95% day-cluster CI --														
IN1	SCR	1.9 dB	vs CA16	6.84 [4.08, 8.94]	vs Caloc	10.95 [7.28,13.95]								
IN4	SCR	1.7 dB	vs CA16	7.07 [4.04, 9.44]	vs Caloc	11.19 [7.42,14.41]								
NA4	SCR	0.5 dB	vs CA16	8.27 [5.76,10.32]	vs Caloc	12.39 [8.88,15.43]								
G1	SCR	1.2 dB	vs CA16	7.52 [5.19, 9.39]	vs Caloc	11.64 [8.28,14.48]								
U	SCR	10.4 dB	vs CA16	-1.64 [-4.25,-0.30]	vs Caloc	2.48 [0.60, 3.74]								
MLP	SCR	6.2 dB	vs CA16	2.55 [1.07, 3.82]	vs Caloc	6.67 [3.97, 8.33]								
LS	SCR	5.1 dB	vs CA16	3.69 [1.34, 5.58]	vs Caloc	7.81 [4.35,10.22]								
-- SCR gain (dB) needed for Pd=0.9; positive = less SCR than reference; 95% day-cluster CI --														
IN1	SCR	11.5 dB	vs CA16	7.22 [4.16, 9.55]	vs Caloc	9.57 [5.97,12.67]								
IN4	SCR	11.2 dB	vs CA16	7.48 [4.09,10.22]	vs Caloc	9.90 [6.03,13.47]								
NA4	SCR	9.7 dB	vs CA16	8.96 [6.01,11.33]	vs Caloc	11.37 [7.84,14.47]								
G1	SCR	10.8 dB	vs CA16	7.88 [5.23,10.04]	vs Caloc	10.22 [6.92,13.23]								
U	SCR	18.6 dB	vs CA16	0.08 [-1.16, 0.98]	vs Caloc	2.35 [0.65, 3.73]								
MLP	SCR	15.9 dB	vs CA16	2.85 [1.26, 4.16]	vs Caloc	5.32 [3.01, 6.76]								
LS	SCR	14.6 dB	vs CA16	4.10 [1.22, 6.30]	vs Caloc	6.51 [2.89, 9.48]								

10 Persistent targets after onset

Frozen onset study (detection/PROTOCOL_ONSET.md): per-look detection probability, cumulative detection within K looks, and clean cumulative false-alarm probabilities:

```
units 56; segments 74980

=== per-look Pfa design 0.01 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1 0.0099 0.0189 0.0357 0.0683
clean cumulative false-alarm prob, K=1,2,4,8: G1 0.0101 0.0192 0.0369 0.0706
clean cumulative false-alarm prob, K=1,2,4,8: IN4 0.0098 0.0191 0.0369 0.0721
clean cumulative false-alarm prob, K=1,2,4,8: NA4 0.0095 0.0182 0.0363 0.0701
clean cumulative false-alarm prob, K=1,2,4,8: CA16 0.0103 0.0170 0.0302 0.0540
clean cumulative false-alarm prob, K=1,2,4,8: CAlc 0.0113 0.0147 0.0209 0.0308
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1 0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002
G1 0.887 0.715 0.645 0.497 0.068 0.011 0.005 0.003
IN4 0.891 0.725 0.563 0.256 0.022 0.003 0.001 0.001
NA4 0.914 0.883 0.705 0.571 0.431 0.287 0.201 0.178
CA16 0.691 0.601 0.455 0.254 0.100 0.039 0.017 0.009
CAlc 0.562 0.564 0.564 0.563 0.565 0.565 0.563 0.565
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001
G1 0.887 0.025 0.014 0.008 0.004 0.003 0.002 0.002
IN4 0.891 0.049 0.004 0.001 0.001 0.001 0.001 0.001
NA4 0.914 0.666 0.153 0.015 0.006 0.004 0.003 0.003
CA16 0.691 0.610 0.460 0.186 0.033 0.013 0.006 0.004
CAlc 0.562 0.563 0.564 0.564 0.564 0.565 0.565 0.565
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002
G1 0.887 0.942 0.909 0.817 0.127 0.014 0.005 0.002
IN4 0.891 0.927 0.763 0.087 0.009 0.002 0.001 0.001
NA4 0.914 0.945 0.931 0.831 0.675 0.500 0.380 0.351
CA16 0.691 0.591 0.454 0.268 0.090 0.040 0.015 0.008
CAlc 0.562 0.564 0.564 0.563 0.564 0.564 0.565 0.562
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1 K=1: 9.82 [ 6.40,12.68] | K=2: 9.92 [ 6.58,12.67] | K=4: 9.19 [ 5.99,11.87] | K=8: 8.42 [ 5.22,11.05]
NA4 K=1: 10.93 [ 7.46,13.74] | K=2: 10.34 [ 7.31,12.81] | K=4: 9.82 [ 7.17,11.99] | K=8: 9.40 [ 7.19,11.28]
IN4 K=1: 10.23 [ 6.71,13.18] | K=2: 9.88 [ 6.44,12.62] | K=4: 9.14 [ 5.82,11.82] | K=8: 8.28 [ 4.83,11.05]
G1 K=1: 9.90 [ 6.58,12.74] | K=2: 9.97 [ 6.75,12.63] | K=4: 9.27 [ 6.23,11.85] | K=8: 8.53 [ 5.56,11.06]
CA16 K=1: 2.76 [ 1.81, 4.06] | K=2: 2.94 [ 1.88, 4.41] | K=4: 2.85 [ 1.67, 4.49] | K=8: 2.45 [ 1.06, 4.25]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1 K=1: 8.95 [ 5.56,11.69] | K=2: 10.00 [ 6.21,12.95] | K=4: 9.96 [ 5.96,13.07] | K=8: 9.45 [ 5.43,12.73]
NA4 K=1: 11.09 [ 7.22,14.10] | K=2: 12.33 [ 7.84,15.68] | K=4: 12.94 [ 8.38,16.44] | K=8: 13.26 [ 8.80,16.82]
IN4 K=1: 9.65 [ 5.85,12.64] | K=2: 10.99 [ 6.43,14.37] | K=4: 10.81 [ 6.09,14.37] | K=8: 10.15 [ 5.18,13.94]
G1 K=1: 9.01 [ 5.66,11.73] | K=2: 10.07 [ 6.39,13.05] | K=4: 10.05 [ 6.23,13.22] | K=8: 9.56 [ 5.56,12.83]
CA16 K=1: 1.41 [ 0.86, 2.14] | K=2: 1.43 [ 0.76, 2.32] | K=4: 1.20 [ 0.38, 2.32] | K=8: 0.53 [-0.47, 1.81]

=== per-look Pfa design 0.001 ===
clean cumulative false-alarm prob, K=1,2,4,8: IN1 0.0011 0.0017 0.0033 0.0063
clean cumulative false-alarm prob, K=1,2,4,8: G1 0.0013 0.0021 0.0040 0.0079
clean cumulative false-alarm prob, K=1,2,4,8: IN4 0.0008 0.0014 0.0027 0.0055
clean cumulative false-alarm prob, K=1,2,4,8: NA4 0.0007 0.0014 0.0027 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CA16 0.0007 0.0015 0.0025 0.0051
clean cumulative false-alarm prob, K=1,2,4,8: CAlc 0.0010 0.0012 0.0018 0.0029
-- per-look Pd at SCR 10 dB, Doppler random (looks j=0..7) --
IN1 0.804 0.587 0.347 0.025 0.001 0.000 0.000 0.000
G1 0.833 0.626 0.455 0.014 0.002 0.000 0.000 0.000
IN4 0.802 0.559 0.133 0.010 0.000 0.000 0.000 0.000
NA4 0.852 0.790 0.516 0.312 0.173 0.124 0.114 0.110
CA16 0.555 0.368 0.125 0.025 0.006 0.002 0.001 0.000
CAlc 0.290 0.290 0.289 0.290 0.289 0.291 0.289 0.289
-- per-look Pd at SCR 10 dB, Doppler matched (looks j=0..7) --
IN1 0.804 0.003 0.000 0.000 0.000 0.000 0.000 0.000
G1 0.833 0.008 0.002 0.001 0.000 0.000 0.000 0.000
IN4 0.802 0.002 0.000 0.000 0.000 0.000 0.000 0.000
NA4 0.852 0.448 0.023 0.001 0.000 0.000 0.000 0.000
CA16 0.555 0.364 0.066 0.004 0.001 0.000 0.000 0.000
CAlc 0.290 0.289 0.291 0.290 0.289 0.289 0.290 0.290
-- per-look Pd at SCR 10 dB, Doppler opposite (looks j=0..7) --
IN1 0.804 0.878 0.629 0.057 0.001 0.000 0.000 0.000
G1 0.833 0.904 0.801 0.025 0.002 0.001 0.000 0.000
IN4 0.802 0.829 0.278 0.001 0.000 0.000 0.000 0.000
NA4 0.852 0.906 0.849 0.540 0.337 0.273 0.259 0.252
CA16 0.555 0.372 0.117 0.032 0.004 0.002 0.000 0.000
CAlc 0.290 0.290 0.291 0.290 0.289 0.292 0.290 0.291
-- SCR (dB) for cumulative Pd=0.5 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1 K=1: 10.93 [ 7.28,14.02] | K=2: 11.99 [ 7.93,15.35] | K=4: 11.48 [ 7.26,15.09] | K=8: 10.88 [ 6.53,14.62]
NA4 K=1: 12.40 [ 9.09,15.50] | K=2: 13.19 [ 9.42,16.38] | K=4: 12.98 [ 9.27,15.90] | K=8: 12.73 [ 9.29,15.59]
IN4 K=1: 11.17 [ 7.43,14.42] | K=2: 12.12 [ 7.77,15.88] | K=4: 11.49 [ 7.08,15.34] | K=8: 10.87 [ 6.35,14.85]
G1 K=1: 11.63 [ 8.41,14.50] | K=2: 12.89 [ 9.11,16.16] | K=4: 12.54 [ 8.65,15.96] | K=8: 11.97 [ 8.03,15.55]
CA16 K=1: 4.06 [ 2.75, 5.93] | K=2: 4.00 [ 2.44, 6.10] | K=4: 3.57 [ 1.85, 5.85] | K=8: 3.00 [ 1.14, 5.43]
-- SCR (dB) for cumulative Pd=0.9 by K, Doppler random; gain vs CAlc [95% day-cluster CI] --
IN1 K=1: 9.54 [ 5.98,12.63] | K=2: 10.64 [ 6.86,13.92] | K=4: 10.38 [ 6.25,13.85] | K=8: 9.80 [ 5.45,13.28]
NA4 K=1: 11.30 [ 7.70,14.40] | K=2: 12.63 [ 8.56,16.01] | K=4: 13.09 [ 9.01,16.52] | K=8: 13.06 [ 9.04,16.55]
IN4 K=1: 9.86 [ 5.87,13.40] | K=2: 11.02 [ 6.57,14.94] | K=4: 10.61 [ 5.84,14.63] | K=8: 10.00 [ 5.16,14.13]
G1 K=1: 10.17 [ 6.99,13.18] | K=2: 11.42 [ 7.90,14.55] | K=4: 11.26 [ 7.42,14.46] | K=8: 10.68 [ 6.87,14.03]
CA16 K=1: 2.45 [ 1.56, 3.81] | K=2: 2.39 [ 1.22, 4.07] | K=4: 2.00 [ 0.67, 3.77] | K=8: 1.40 [-0.11, 3.33]
```

Exploratory: comparison at equal dwell false-alarm probability (0.01), with max and sum-of-squares dwell statistics:

```
EXPLORATORY: units 56; dwell Pfa design 0.01; persistent Swerling-1 targets, random Doppler

-- K = 1 looks --
IN1 max: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
IN1 sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -0.9 dB | Pd 0.9 8.8 dB
NA4 max: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
NA4 sum: measured dwell Pfa 0.0095 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 6.6 dB
CAlc max: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
CAlc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 9.0 dB | Pd 0.9 17.8 dB
gain of IN1-max over integrated power CFAR (CAlc-sum) at Pd 0.5: 9.93 [ 6.46, 12.81] dB
gain of IN1-max over integrated power CFAR (CAlc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-max over integrated power CFAR (CAlc-sum) at Pd 0.5: 10.99 [ 7.58, 13.81] dB
gain of NA4-max over integrated power CFAR (CAlc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB
gain of IN1-sum over integrated power CFAR (CAlc-sum) at Pd 0.5: 9.93 [ 6.37, 12.81] dB
gain of IN1-sum over integrated power CFAR (CAlc-sum) at Pd 0.9: 8.94 [ 5.54, 11.59] dB
gain of NA4-sum over integrated power CFAR (CAlc-sum) at Pd 0.5: 10.99 [ 7.61, 13.81] dB
gain of NA4-sum over integrated power CFAR (CAlc-sum) at Pd 0.9: 11.20 [ 7.46, 14.19] dB

-- K = 2 looks --
IN1 max: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.0 dB | Pd 0.9 7.1 dB
IN1 sum: measured dwell Pfa 0.0097 | SCR for Pd 0.5 -2.2 dB | Pd 0.9 6.8 dB
NA4 max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -2.5 dB | Pd 0.9 4.8 dB
```

```

NA4  sum: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
CAloc max: measured dwell Pfa 0.0110 | SCR for Pd 0.5 8.1 dB | Pd 0.9 16.8 dB
CAloc sum: measured dwell Pfa 0.0108 | SCR for Pd 0.5 8.7 dB | Pd 0.9 17.2 dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.72 [ 6.85, 13.68] dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 10.12 [ 6.29, 13.09] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.15 [ 7.71, 13.85] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 12.46 [ 7.90, 15.85] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.91 [ 7.32, 13.74] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 10.40 [ 6.73, 13.29] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.42 [ 8.19, 13.92] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 13.13 [ 8.71, 16.43] dB

-- K = 4 looks --
IN1  max: measured dwell Pfa 0.0094 | SCR for Pd 0.5 -1.9 dB | Pd 0.9 7.4 dB
IN1  sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -2.6 dB | Pd 0.9 6.6 dB
NA4  max: measured dwell Pfa 0.0087 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 4.1 dB
NA4  sum: measured dwell Pfa 0.0099 | SCR for Pd 0.5 -3.3 dB | Pd 0.9 2.7 dB
CAloc max: measured dwell Pfa 0.0109 | SCR for Pd 0.5 7.6 dB | Pd 0.9 16.2 dB
CAloc sum: measured dwell Pfa 0.0111 | SCR for Pd 0.5 8.3 dB | Pd 0.9 16.8 dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.29 [ 6.26, 13.55] dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 9.47 [ 5.33, 12.64] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.08 [ 7.57, 13.82] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 12.70 [ 7.98, 16.26] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.94 [ 7.48, 13.70] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 10.22 [ 6.66, 13.08] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.67 [ 8.74, 13.95] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 14.17 [ 9.70, 17.61] dB

-- K = 8 looks --
IN1  max: measured dwell Pfa 0.0093 | SCR for Pd 0.5 -1.3 dB | Pd 0.9 8.3 dB
IN1  sum: measured dwell Pfa 0.0102 | SCR for Pd 0.5 -2.7 dB | Pd 0.9 7.0 dB
NA4  max: measured dwell Pfa 0.0091 | SCR for Pd 0.5 -2.9 dB | Pd 0.9 4.3 dB
NA4  sum: measured dwell Pfa 0.0100 | SCR for Pd 0.5 -3.8 dB | Pd 0.9 2.4 dB
CAloc max: measured dwell Pfa 0.0103 | SCR for Pd 0.5 7.4 dB | Pd 0.9 16.0 dB
CAloc sum: measured dwell Pfa 0.0105 | SCR for Pd 0.5 8.0 dB | Pd 0.9 16.4 dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 9.31 [ 4.87, 12.82] dB
gain of IN1-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 8.14 [ 3.83, 11.48] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.85 [ 7.29, 13.54] dB
gain of NA4-max over integrated power CFAR (CAloc-sum) at Pd 0.9: 12.13 [ 7.50, 15.64] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 10.70 [ 7.35, 13.40] dB
gain of IN1-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 9.35 [ 6.02, 12.16] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.5: 11.74 [ 9.17, 13.75] dB
gain of NA4-sum over integrated power CFAR (CAloc-sum) at Pd 0.9: 14.01 [ 9.97, 17.21] dB

```

Exploratory: exact-law prediction of the post-onset decay:

```

Exact-law prediction vs measured per-look Pd after onset (IN-ARCP, SCR 10 dB, Pfa 0.01); pooled over 56 units (exploratory)
random  predicted 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
        measured 0.885 0.711 0.639 0.478 0.084 0.012 0.005 0.002   MAE per unit x look 0.010
matched predicted 0.885 0.023 0.013 0.007 0.004 0.002 0.002 0.001
        measured 0.885 0.024 0.013 0.007 0.003 0.002 0.002 0.001   MAE per unit x look 0.003
opposite predicted 0.885 0.941 0.906 0.793 0.167 0.014 0.004 0.002
        measured 0.885 0.940 0.905 0.796 0.165 0.014 0.004 0.002   MAE per unit x look 0.008

```

11 Numerical checks of the detection theory

Monte Carlo checks of Corollary 2, Proposition 2 (closed forms) and Proposition 3 (output of `detection/theory_checks.py`):

```

T1/T2 exact detection law vs Monte Carlo (Swirling-1 target in Y only; 400k draws)
rho=0.91-0.18j c=None r=0.91-0.18j SCR=1.0: MC 0.5358 exact 0.5364 closed form 0.5364
rho=0.91-0.18j c=10.0 r=0.88-0.18j SCR=3.0: MC 0.6381 exact 0.6374
rho=0.70 c=1.0 r=0.65 SCR=10.0: MC 0.4275 exact 0.4263

T3 noise-limited whitening gain: empirical (c+1)/E|Y-pred|^2 vs closed forms (m=16 history, 200k draws)
|rho|=0.7 c= 0.1: IN emp -1.46 dB closed -1.46 | oracle emp 0.03 dB closed(m=inf) 0.03 | clutter-only bound 2.92
|rho|=0.7 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 0.66 dB closed(m=inf) 0.67 | clutter-only bound 2.92
|rho|=0.7 c= 10.0: IN emp 2.24 dB closed 2.23 | oracle emp 2.29 dB closed(m=inf) 2.27 | clutter-only bound 2.92
|rho|=0.7 c= 100.0: IN emp 2.85 dB closed 2.84 | oracle emp 2.85 dB closed(m=inf) 2.84 | clutter-only bound 2.92
|rho|=0.93 c= 0.1: IN emp -2.32 dB closed -2.32 | oracle emp 0.12 dB closed(m=inf) 0.12 | clutter-only bound 8.69
|rho|=0.93 c= 1.0: IN emp 0.02 dB closed 0.00 | oracle emp 1.66 dB closed(m=inf) 1.65 | clutter-only bound 8.69
|rho|=0.93 c= 10.0: IN emp 5.34 dB closed 5.34 | oracle emp 5.76 dB closed(m=inf) 5.76 | clutter-only bound 8.69
|rho|=0.93 c= 100.0: IN emp 8.17 dB closed 8.18 | oracle emp 8.18 dB closed(m=inf) 8.19 | clutter-only bound 8.69
|rho|=0.98 c= 0.1: IN emp -2.50 dB closed -2.52 | oracle emp 0.20 dB closed(m=inf) 0.21 | clutter-only bound 14.02
|rho|=0.98 c= 1.0: IN emp -0.01 dB closed 0.00 | oracle emp 2.22 dB closed(m=inf) 2.22 | clutter-only bound 14.02
|rho|=0.98 c= 10.0: IN emp 6.69 dB closed 6.69 | oracle emp 7.78 dB closed(m=inf) 7.78 | clutter-only bound 14.02
|rho|=0.98 c= 100.0: IN emp 12.31 dB closed 12.32 | oracle emp 12.43 dB closed(m=inf) 12.44 | clutter-only bound 14.02

T4 persistent strong target: limiting score and Pd (IN-ARCP, r fitted = 0.93e^{-0.2j})
sqrt(m/(m-1)) = 1.033; thresholds q(0.1)=1.574 q(0.01)=2.310
Doppler w=+0.0: limiting score 0.937 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=-0.2: limiting score 0.613 | SCR 0 dB: Pd(0.01) 0.010 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000
Doppler w=+1.0: limiting score 1.029 | SCR 0 dB: Pd(0.01) 0.000 | SCR 20 dB: Pd(0.01) 0.000 | SCR 40 dB: Pd(0.01) 0.000

```

12 Baselines: parametric texture and adaptive conformal inference

B1/B2 within-session units: 56; transfer days: 6

```

=== 1-alpha = 0.9 ===
B1 NA4-K / NA4 radius (GM): 1.0092 [1.0064, 1.0115] | coverage NA4 0.9019 NA4-K 0.9015 | max dev NA4 0.043 NA4-K 0.043
B2 within ACI g=0.005: coverage 0.9010 vs split 0.9013 | radius ratio (finite) 0.999 [0.997,1.000] | infinite-region fraction 0.0000 | max dev 0.033 vs 0.036
B2 within ACI g=0.02: coverage 0.9007 vs split 0.9013 | radius ratio (finite) 0.999 [0.995,1.002] | infinite-region fraction 0.0000 | max dev 0.029 vs 0.036
B2 transfer IN1 : mean coverage 0.9029 worst day 0.8822 | radius vs split 1.000 | max dev 0.024 | per day 0.905 0.894 0.915 0.918 0.904 0.882
B2 transfer ACIO.005 : mean coverage 0.9032 worst day 0.8889 | radius vs split 1.001 | max dev 0.020 | per day 0.904 0.897 0.912 0.914 0.903 0.889
B2 transfer ACIO.02 : mean coverage 0.9024 worst day 0.8965 | radius vs split 1.000 | max dev 0.015 | per day 0.903 0.900 0.906 0.908 0.902 0.897

=== 1-alpha = 0.99 ===
B1 NA4-K / NA4 radius (GM): 1.0078 [1.0034, 1.0104] | coverage NA4 0.9905 NA4-K 0.9906 | max dev NA4 0.011 NA4-K 0.011
B2 within ACI g=0.005: coverage 0.9903 vs split 0.9901 | radius ratio (finite) 1.001 [0.996,1.007] | infinite-region fraction 0.0000 | max dev 0.010 vs 0.011
B2 within ACI g=0.02: coverage 0.9901 vs split 0.9901 | radius ratio (finite) 1.006 [0.998,1.015] | infinite-region fraction 0.0115 | max dev 0.009 vs 0.011
B2 transfer IN1 : mean coverage 0.9921 worst day 0.9875 | radius vs split 1.000 | max dev 0.006 | per day 0.987 0.992 0.994 0.996 0.994 0.989
B2 transfer ACIO.005 : mean coverage 0.9918 worst day 0.9889 | radius vs split 0.997 | max dev 0.005 | per day 0.989 0.991 0.993 0.995 0.993 0.990
B2 transfer ACIO.02 : mean coverage 0.9912 worst day 0.9902 | radius vs split 0.995 | max dev 0.004 | per day 0.990 0.990 0.992 0.993 0.991 0.990

```

13 Classical detectors, gradual emergence and ACI contamination (R10)

Frozen protocol r10/PROTOCOL_R10.md; stated expectations and verdicts in r10/RESULTS_R10.md. Output of r10/analyze_r10.py:

```
units 56; days 6; test segments 74980

=== Part C: dwell detectors at equal dwell Pfa (alpha = 0.01 unless stated) ===
-- measured dwell Pfa on clean test segments, alpha = 0.01 (K' = 1 2 4 8) --
IN1sum 0.0099 0.0097 0.0099 0.0102
NA4sum 0.0095 0.0094 0.0099 0.0100
CALocsum 0.0108 0.0108 0.0111 0.0105
PAMFH 0.0098 0.0109 0.0101 0.0096
NPAMF 0.0098 0.0109 0.0098 0.0101
MTDhann nan 0.0095 0.0095 0.0106
MTDrect nan 0.0095 0.0096 0.0097
-- measured dwell Pfa on clean test segments, alpha = 0.001 (K' = 1 2 4 8) --
IN1sum 0.0011 0.0008 0.0008 0.0009
NA4sum 0.0007 0.0011 0.0009 0.0010
CALocsum 0.0009 0.0010 0.0011 0.0012
PAMFH 0.0008 0.0010 0.0009 0.0010
NPAMF 0.0008 0.0009 0.0009 0.0012
MTDhann nan 0.0005 0.0009 0.0008
MTDrect nan 0.0005 0.0010 0.0009
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler random, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 -2.2/ 6.7 -2.5/ 6.5 -2.7/ 6.9
NA4sum -1.9/ 6.6 -2.7/ 4.1 -3.3/ 2.7 -3.7/ 2.3
CALocsum 9.0/ 17.7 8.7/ 17.2 8.4/ 16.8 8.0/ 16.4
PAMFH -1.3/ 8.0 -2.4/ 5.1 -3.1/ 3.7 -3.7/ 3.0
NPAMF -1.3/ 8.0 -2.4/ 5.1 -3.1/ 3.6 -3.8/ 2.6
MTDhann - 8.4/ 20.7 4.5/ 19.2 2.3/ 17.9
MTDrect - 8.4/ 20.7 7.7/ 19.9 6.3/ 18.9
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler random, alpha 0.001; mean over units --
IN1sum 2.0/ 11.5 0.3/ 10.0 0.4/ 10.3 1.0/ 11.3
NA4sum 0.5/ 9.8 -1.3/ 7.5 -1.3/ 6.8 -1.4/ 7.2
CALocsum 13.1/ 21.0 12.8/ 20.5 12.6/ 20.2 12.2/ 20.1
PAMFH 1.8/ 11.2 -0.4/ 8.9 -1.2/ 7.6 -2.0/ 7.0
NPAMF 1.8/ 11.2 -0.6/ 8.6 -2.0/ 6.4 -3.0/ 5.0
MTDhann - 14.8/ 22.4 11.9/ 22.5 10.6/ 22.2
MTDrect - 14.8/ 22.4 14.9/ 22.6 13.8/ 22.1
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler matched, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 0.3/ 10.0 2.1/ 11.6 4.0/ 13.4
NA4sum -1.9/ 6.6 -2.0/ 6.7 -1.5/ 7.5 -0.1/ 9.2
CALocsum 9.0/ 17.7 8.8/ 17.5 8.5/ 17.2 8.2/ 16.8
PAMFH -1.3/ 8.0 -1.0/ 8.4 -0.3/ 9.1 0.6/ 10.1
NPAMF -1.3/ 8.0 -1.0/ 8.4 -0.3/ 9.0 0.2/ 9.5
MTDhann - 18.3/ 22.1 18.3/ 21.6 19.0/ 20.2
MTDrect - 18.3/ 22.1 18.3/ 21.4 17.8/ 21.7
-- SCR (dB) for dwell Pd = 0.5 / 0.9, Doppler opposite, alpha 0.01; mean over units --
IN1sum -0.8/ 8.7 -3.0/ 3.6 -3.6/ 1.9 -3.8/ 1.3
NA4sum -1.9/ 6.6 -3.1/ 1.9 -3.7/ 0.3 -4.1/ -0.8
CALocsum 9.0/ 17.7 8.5/ 16.9 8.3/ 16.6 7.9/ 16.3
PAMFH -1.3/ 8.0 -3.0/ 2.2 -3.6/ 0.1 -4.1/ -1.1
NPAMF -1.3/ 8.0 -3.0/ 2.2 -3.6/ -0.1 -4.2/ -1.3
MTDhann - 4.5/ 15.1 0.3/ 9.4 -0.5/ 8.5
MTDrect - 4.5/ 15.1 4.0/ 14.6 2.6/ 13.0
-- gains (dB, positive = first detector needs less SCR) at Pd = 0.5, random Doppler, alpha 0.01; 95% day bootstrap --
K'=1 Pd=0.5 IN1sum vs PAMFH : -0.43 [ -0.59, -0.24] LOD0 -0.48..-0.37
K'=1 Pd=0.5 IN1sum vs PAMFH : -0.72 [ -1.07, -0.26] LOD0 -0.86..-0.44
K'=1 Pd=0.5 NA4sum vs PAMFH : 0.66 [ 0.44, 0.91] LOD0 0.59.. 0.75
K'=1 Pd=0.5 NA4sum vs PAMFH : 1.43 [ 1.37, 1.51] LOD0 1.41.. 1.45
K'=1 Pd=0.5 IN1sum vs CALocsum: 9.86 [ 6.31, 12.74] LOD0 8.99..11.60
K'=1 Pd=0.5 IN1sum vs CALocsum: 9.00 [ 5.58, 11.67] LOD0 8.07..10.65
K'=1 Pd=0.5 PAMFH vs CALocsum: 10.29 [ 6.79, 13.25] LOD0 9.39..12.07
K'=1 Pd=0.5 PAMFH vs CALocsum: 9.72 [ 6.08, 12.63] LOD0 8.51..11.52
K'=1 Pd=0.5 PAMFH vs NPAMF : 0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
K'=1 Pd=0.5 PAMFH vs NPAMF : 0.00 [ 0.00, 0.00] LOD0 0.00.. 0.00
K'=2 Pd=0.5 IN1sum vs PAMFH : -0.20 [ -0.30, 0.00] LOD0 -0.24..-0.15
K'=2 Pd=0.5 IN1sum vs PAMFH : -1.68 [ -2.39, -0.77] LOD0 -1.99..-1.10
K'=2 Pd=0.5 NA4sum vs PAMFH : 0.32 [ 0.07, 0.67] LOD0 0.17.. 0.47
K'=2 Pd=0.5 NA4sum vs PAMFH : 1.00 [ 0.84, 1.19] LOD0 0.93.. 1.13
K'=2 Pd=0.5 IN1sum vs MTDhann: 10.59 [ 9.46, 12.27] LOD0 10.19..11.13
K'=2 Pd=0.5 IN1sum vs MTDhann: 14.19 [ 12.11, 16.48] LOD0 13.70..15.24
K'=2 Pd=0.5 NA4sum vs MTDhann: 11.11 [ 10.34, 12.46] LOD0 10.81..11.45
K'=2 Pd=0.5 NA4sum vs MTDhann: 16.88 [ 14.09, 19.13] LOD0 16.31..18.18
K'=2 Pd=0.5 PAMFH vs MTDhann: 10.79 [ 9.75, 12.25] LOD0 10.43..11.28
K'=2 Pd=0.5 PAMFH vs MTDhann: 15.92 [ 12.95, 18.05] LOD0 15.25..17.25
K'=2 Pd=0.5 IN1sum vs CALocsum: 10.84 [ 7.30, 13.75] LOD0 9.98..12.63
K'=2 Pd=0.5 IN1sum vs CALocsum: 10.48 [ 6.68, 13.88] LOD0 9.42..12.33
K'=2 Pd=0.5 PAMFH vs CALocsum: 11.04 [ 7.55, 13.80] LOD0 10.13..12.79
K'=2 Pd=0.5 PAMFH vs CALocsum: 12.16 [ 7.57, 15.49] LOD0 10.53..14.32
K'=2 Pd=0.5 MTDhann vs CALocsum: 0.26 [ -2.49, 2.31] LOD0 -0.84.. 1.50
K'=2 Pd=0.5 MTDhann vs CALocsum: -3.47 [ -5.49, -2.08] LOD0 -4.46..-2.73
K'=2 Pd=0.5 PAMFH vs NPAMF : 0.00 [ -0.00, 0.00] LOD0 0.00.. 0.00
K'=2 Pd=0.5 PAMFH vs NPAMF : 0.01 [ -0.03, 0.04] LOD0 -0.00.. 0.02
K'=4 Pd=0.5 IN1sum vs PAMFH : -0.54 [ -0.88, -0.19] LOD0 -0.63..-0.37
K'=4 Pd=0.5 IN1sum vs PAMFH : -2.78 [ -3.49, -1.87] LOD0 -3.11..-2.23
K'=4 Pd=0.5 NA4sum vs PAMFH : 0.22 [ 0.04, 0.47] LOD0 0.10.. 0.29
K'=4 Pd=0.5 NA4sum vs PAMFH : 1.00 [ 0.81, 1.27] LOD0 0.91.. 1.11
K'=4 Pd=0.5 IN1sum vs MTDhann: 6.98 [ 5.95, 8.43] LOD0 6.74.. 7.86
K'=4 Pd=0.5 IN1sum vs MTDhann: 12.84 [ 11.08, 14.97] LOD0 12.34..13.68
K'=4 Pd=0.5 NA4sum vs MTDhann: 7.74 [ 6.53, 9.23] LOD0 7.35.. 8.73
K'=4 Pd=0.5 NA4sum vs MTDhann: 16.63 [ 14.09, 18.91] LOD0 15.98..17.77
K'=4 Pd=0.5 PAMFH vs MTDhann: 7.52 [ 6.45, 8.90] LOD0 7.25.. 8.45
K'=4 Pd=0.5 PAMFH vs MTDhann: 15.68 [ 13.18, 17.78] LOD0 15.12..16.79
K'=4 Pd=0.5 IN1sum vs CALocsum: 10.88 [ 7.44, 13.64] LOD0 10.04..12.60
K'=4 Pd=0.5 IN1sum vs CALocsum: 10.32 [ 6.71, 13.19] LOD0 9.41..12.08
K'=4 Pd=0.5 PAMFH vs CALocsum: 11.42 [ 8.29, 13.88] LOD0 10.63..12.98
K'=4 Pd=0.5 PAMFH vs CALocsum: 13.10 [ 8.63, 16.42] LOD0 11.64..15.19
K'=4 Pd=0.5 MTDhann vs CALocsum: 3.90 [ -0.24, 6.88] LOD0 2.19.. 5.72
K'=4 Pd=0.5 MTDhann vs CALocsum: -2.39 [ -4.54, -0.89] LOD0 -3.36..-1.60
K'=4 Pd=0.5 PAMFH vs NPAMF : -0.04 [ -0.11, 0.01] LOD0 -0.06..-0.00
K'=4 Pd=0.5 PAMFH vs NPAMF : -0.13 [ -0.21, -0.08] LOD0 -0.16..-0.11
K'=8 Pd=0.5 IN1sum vs PAMFH : -1.01 [ -1.67, -0.34] LOD0 -1.18..-0.70
K'=8 Pd=0.5 IN1sum vs PAMFH : -3.95 [ -4.43, -3.32] LOD0 -4.17..-3.57
K'=8 Pd=0.5 NA4sum vs PAMFH : 0.08 [ -0.14, 0.30] LOD0 -0.04.. 0.14
K'=8 Pd=0.5 NA4sum vs PAMFH : 0.70 [ 0.49, 1.03] LOD0 0.62.. 0.77
K'=8 Pd=0.5 IN1sum vs MTDhann: 4.95 [ 4.19, 5.97] LOD0 4.75.. 5.58
K'=8 Pd=0.5 IN1sum vs MTDhann: 10.96 [ 9.11, 12.61] LOD0 10.54..11.83
K'=8 Pd=0.5 NA4sum vs MTDhann: 6.04 [ 4.85, 7.58] LOD0 5.41.. 6.80
K'=8 Pd=0.5 NA4sum vs MTDhann: 15.60 [ 13.19, 17.56] LOD0 14.90..16.71
K'=8 Pd=0.5 PAMFH vs MTDhann: 5.96 [ 4.85, 7.48] LOD0 5.45.. 6.71
K'=8 Pd=0.5 PAMFH vs MTDhann: 14.90 [ 12.56, 16.68] LOD0 14.14..15.99
K'=8 Pd=0.5 IN1sum vs CALocsum: 10.65 [ 7.32, 13.37] LOD0 9.82..12.34
K'=8 Pd=0.5 IN1sum vs CALocsum: 9.41 [ 5.89, 12.28] LOD0 8.54..11.12
K'=8 Pd=0.5 PAMFH vs CALocsum: 11.66 [ 8.91, 13.74] LOD0 10.94..13.03
```

```

K'=8 Pd=0.9 PAMFH vs Calocsum: 13.36 [ 9.35, 16.42] LOD0 12.11..15.29
K'=8 Pd=0.5 MTDhann vs Calocsum: 5.70 [ 1.71, 8.63] LOD0 4.24.. 7.58
K'=8 Pd=0.9 MTDhann vs Calocsum: -1.54 [ -3.31, -0.10] LOD0 -2.03..-0.71
K'=8 Pd=0.5 PAMFH vs NPAMF : -0.15 [ -0.27, -0.04] LOD0 -0.18..-0.09
K'=8 Pd=0.9 PAMFH vs NPAMF : -0.44 [ -0.55, -0.30] LOD0 -0.48..-0.40

C1 check: max |Pd(PAMF-H, K'=1) - Pd(IN4 look 0)| over SCR = 0.0000 (expectation < 0.005)
C2: PAMF-H needs 1.01 dB less SCR than IN1-sum at K'=8 (expectation >= 3 dB): FAILED
C3: MTD-CA needs 5.70 dB less SCR than Caloc-sum at K'=8 (expectation >= 5 dB): OK
C4: PAMF-H matched-Doppler SCR is 4.25 dB worse than random (expectation >= 3 dB): OK

=== Part R: gradual emergence (random Doppler, alpha 0.01) ===
-- placement pre: per-look Pd at SCR 10 dB (looks 0..7), mean over units --
L= 1 IN1 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
L= 1 NA4 0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
L= 1 Caloc 0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
L= 4 IN1 0.793 0.705 0.549 0.099 0.015 0.006 0.003 0.002
L= 4 NA4 0.790 0.821 0.643 0.473 0.281 0.203 0.183 0.173
L= 4 Caloc 0.559 0.556 0.557 0.557 0.559 0.558 0.559 0.557
L= 8 IN1 0.656 0.446 0.047 0.012 0.005 0.003 0.002 0.001
L= 8 NA4 0.678 0.684 0.404 0.237 0.194 0.179 0.171 0.166
L= 8 Caloc 0.557 0.559 0.558 0.556 0.558 0.557 0.558 0.557
L=16 IN1 0.014 0.007 0.003 0.002 0.001 0.001 0.001 0.001
L=16 NA4 0.204 0.203 0.184 0.175 0.168 0.165 0.162 0.160
L=16 Caloc 0.558 0.558 0.557 0.557 0.558 0.558 0.560 0.559
-- placement pre: SCR for Pd 0.5 (single look 0 | any of 8 looks) and first-detection look at 10 dB --
L= 1 IN1 look0 -0.8 | cum8 -3.1 | first look 0.12
L= 1 IN4 look0 -1.3 | cum8 -3.0 | first look 0.10
L= 1 NA4 look0 -1.9 | cum8 -4.1 | first look 0.14
L= 1 CA16 look0 6.2 | cum8 2.8 | first look 0.27
L= 1 Caloc look0 9.0 | cum8 5.4 | first look 0.63
L= 4 IN1 look0 0.4 | cum8 -2.5 | first look 0.16
L= 4 IN4 look0 0.1 | cum8 -1.9 | first look 0.12
L= 4 NA4 look0 -1.7 | cum8 -4.0 | first look 0.22
L= 4 CA16 look0 7.7 | cum8 4.2 | first look 0.32
L= 4 Caloc look0 9.0 | cum8 5.4 | first look 0.63
L= 8 IN1 look0 4.4 | cum8 0.3 | first look 0.21
L= 8 IN4 look0 9.3 | cum8 7.4 | first look 0.21
L= 8 NA4 look0 -0.9 | cum8 -4.0 | first look 0.27
L= 8 CA16 look0 11.7 | cum8 8.3 | first look 0.42
L= 8 Caloc look0 9.1 | cum8 5.4 | first look 0.63
L=16 IN1 look0 nan | cum8 nan | first look 1.49
L=16 IN4 look0 nan | cum8 nan | first look 2.29
L=16 NA4 look0 18.7 | cum8 13.0 | first look 0.67
L=16 CA16 look0 nan | cum8 nan | first look 0.88
L=16 Caloc look0 9.1 | cum8 5.4 | first look 0.63
-- placement pre: SCR for dwell Pd 0.5 at K=8 (dwell detectors) --
L= 1 IN1sum -2.7 NA4sum -3.7 Calocsum 8.0 PAMFH -3.7 NPAMF -3.8 MTDhann 2.3 MTDirect 6.3
L= 4 IN1sum -2.1 NA4sum -3.6 Calocsum 8.0 PAMFH -3.5 NPAMF -3.6 MTDhann 2.3 MTDirect 6.2
L= 8 IN1sum 0.3 NA4sum -3.5 Calocsum 8.0 PAMFH -3.1 NPAMF -3.5 MTDhann 2.3 MTDirect 6.3
L=16 IN1sum nan NA4sum 9.3 Calocsum 8.0 PAMFH -1.8 NPAMF -3.0 MTDhann 2.3 MTDirect 6.2
-- placement in: per-look Pd at SCR 10 dB (looks 0..7), mean over units --
L= 1 IN1 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
L= 1 NA4 0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
L= 1 Caloc 0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
L= 4 IN1 0.465 0.676 0.757 0.794 0.703 0.547 0.097 0.015
L= 4 NA4 0.558 0.726 0.786 0.792 0.821 0.643 0.471 0.280
L= 4 Caloc 0.036 0.179 0.388 0.569 0.557 0.558 0.558 0.557
L= 8 IN1 0.212 0.443 0.555 0.617 0.651 0.670 0.673 0.656
L= 8 NA4 0.304 0.529 0.630 0.677 0.695 0.701 0.693 0.677
L= 8 Caloc 0.015 0.037 0.092 0.180 0.285 0.387 0.479 0.557
L=16 IN1 0.065 0.217 0.333 0.411 0.456 0.482 0.491 0.484
L=16 NA4 0.104 0.307 0.436 0.513 0.555 0.579 0.585 0.583
L=16 Caloc 0.012 0.016 0.023 0.036 0.059 0.091 0.133 0.179
-- placement in: SCR for Pd 0.5 (single look 0 | any of 8 looks) and first-detection look at 10 dB --
L= 1 IN1 look0 -0.8 | cum8 -3.1 | first look 0.12
L= 1 IN4 look0 -1.3 | cum8 -3.0 | first look 0.10
L= 1 NA4 look0 -1.9 | cum8 -4.1 | first look 0.14
L= 1 CA16 look0 6.2 | cum8 2.8 | first look 0.27
L= 1 Caloc look0 9.0 | cum8 5.4 | first look 0.63
L= 4 IN1 look0 11.0 | cum8 -2.8 | first look 0.91
L= 4 IN4 look0 10.2 | cum8 -2.5 | first look 0.82
L= 4 NA4 look0 8.9 | cum8 -3.8 | first look 0.77
L= 4 CA16 look0 18.2 | cum8 3.7 | first look 1.70
L= 4 Caloc look0 21.0 | cum8 6.0 | first look 2.46
L= 8 IN1 look0 16.2 | cum8 -2.0 | first look 1.94
L= 8 IN4 look0 15.4 | cum8 -1.6 | first look 1.73
L= 8 NA4 look0 14.3 | cum8 -3.0 | first look 1.60
L= 8 CA16 look0 23.6 | cum8 5.6 | first look 3.23
L= 8 Caloc look0 24.8 | cum8 7.5 | first look 4.35
L=16 IN1 look0 19.9 | cum8 3.1 | first look 2.84
L=16 IN4 look0 18.9 | cum8 2.9 | first look 2.58
L=16 NA4 look0 17.7 | cum8 -0.2 | first look 2.54
L=16 CA16 look0 nan | cum8 11.6 | first look 4.01
L=16 Caloc look0 nan | cum8 13.6 | first look 4.99
-- placement in: SCR for dwell Pd 0.5 at K=8 (dwell detectors) --
L= 1 IN1sum -2.7 NA4sum -3.7 Calocsum 8.0 PAMFH -3.7 NPAMF -3.8 MTDhann 2.3 MTDirect 6.3
L= 4 IN1sum -2.1 NA4sum -3.2 Calocsum 9.3 PAMFH -3.0 NPAMF -3.1 MTDhann 2.9 MTDirect 7.7
L= 8 IN1sum -0.3 NA4sum -2.3 Calocsum 12.0 PAMFH -1.8 NPAMF -1.7 MTDhann 6.8 MTDirect 10.7
L=16 IN1sum 5.6 NA4sum 1.7 Calocsum 18.0 PAMFH 2.8 NPAMF 3.2 MTDhann 12.9 MTDirect 16.8
R-pre single-look gain IN1 vs Caloc, L= 1: 9.86 [ 6.31, 12.78] LOD0 8.99..11.60
R-pre single-look gain IN1 vs Caloc, L= 4: 8.65 [ 5.11, 11.50] LOD0 7.82..10.40
R-pre single-look gain IN1 vs Caloc, L= 8: 4.65 [ 0.90, 7.62] LOD0 3.71.. 6.54
R-pre single-look gain IN1 vs Caloc, L=16: nan [ nan, nan] LOD0 nan.. nan
R1: gain decreasing in L: False; below 5 dB at L=16: False -> FAILED
R3: IN1 first-detection look (R-in) by L: [0.12, 0.91, 1.94, 2.84] -> OK

=== Part A: ACI with target-contaminated feedback (IN1, alpha 0.01) ===
pi=0 split Pfa/a 0.99 Pd nan | naive0.005 Pfa/a 0.97 Pd nan | gated0.005 Pfa/a 0.97 Pd nan | naive0.02 Pfa/a 0.99 Pd nan | gated0.02 Pfa/a 0.99 Pd nan
pi=0.01 split Pfa/a 0.99 Pd 0.893 | naive0.005 Pfa/a 0.68 Pd 0.880 | gated0.005 Pfa/a 0.89 Pd 0.888 | naive0.02 Pfa/a 0.52 Pd 0.664 | gated0.02 Pfa/a 0.82 Pd 0.850
pi=0.05 split Pfa/a 0.99 Pd 0.890 | naive0.005 Pfa/a 0.24 Pd 0.426 | gated0.005 Pfa/a 0.68 Pd 0.834 | naive0.02 Pfa/a 0.14 Pd 0.275 | gated0.02 Pfa/a 0.55 Pd 0.635
A1 (gamma 0.02): naive clean Pfa/alpha 0.14 (< 0.87) and Pd 0.275 vs split 0.890 -> OK
A2 (gamma 0.02): gated clean Pfa/alpha 0.55 within [0.8, 1.2]? -> FAILED

```

Exact-law prediction of IN-ARCP's per-look detection probability under gradual emergence (endpoint R2):

```

pre|L1 pred 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
meas 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002 | MAE 0.009
pre|L4 pred 0.789 0.702 0.541 0.101 0.015 0.006 0.003 0.002
meas 0.793 0.705 0.549 0.099 0.015 0.006 0.003 0.002 | MAE 0.009
pre|L8 pred 0.648 0.434 0.049 0.012 0.005 0.003 0.002 0.001
meas 0.656 0.446 0.047 0.012 0.005 0.003 0.002 0.001 | MAE 0.008
pre|L16 pred 0.015 0.007 0.004 0.002 0.002 0.001 0.001 0.001
meas 0.014 0.007 0.003 0.002 0.001 0.001 0.001 0.001 | MAE 0.001
in|L1 pred 0.885 0.710 0.635 0.472 0.086 0.012 0.004 0.002
meas 0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002 | MAE 0.009

```

```

in|L4    pred 0.429 0.665 0.750 0.789 0.702 0.541 0.101 0.015
        meas 0.465 0.676 0.757 0.794 0.703 0.547 0.097 0.015 | MAE 0.021
in|L8    pred 0.155 0.407 0.535 0.604 0.643 0.662 0.665 0.648
        meas 0.212 0.443 0.555 0.617 0.651 0.670 0.673 0.656 | MAE 0.033
in|L16   pred 0.038 0.168 0.296 0.379 0.432 0.461 0.472 0.462
        meas 0.065 0.217 0.333 0.411 0.456 0.482 0.491 0.484 | MAE 0.037
R2: exact-law MAE per unit x look x L x placement = 0.016 (expectation <= 0.03): OK

```

14 Order-statistic normalization and real interference (R11)

Frozen protocol r11/PROTOCOL.R11.md; expectations and verdicts in r11/RESULTS.R11.md (2 of 11 met). Monte Carlo check of Corollary 3 (r11/theory_os.py):

```

Pfa T=1.0: law 0.29412 MC 0.29393 z=-0.26
Pfa T=3.0: law 0.03612 MC 0.03613 z=+0.03
Pfa T=8.0: law 0.00067 MC 0.00066 z=-0.26
Pd S=1.0: law 0.5160 MC 0.5166
Pd S=10.0: law 0.9266 MC 0.9269
OS(k=12) loss vs CA in homogeneous clutter at Pfa 0.01, Pd 0.5: 0.30 dB
0 contaminated history samples (random positions): Pd at S=10 OS 0.926 RMS 0.931
2 contaminated history samples (random positions): Pd at S=10 OS 0.835 RMS 0.000
4 contaminated history samples (random positions): Pd at S=10 OS 0.000 RMS 0.000
5 contaminated history samples (random positions): Pd at S=10 OS 0.000 RMS 0.000

```

Output of r11/analyze_r11.py (Part O: IPIX; Part J: 77 GHz with real interference; N: no zeroing, Z: fast-time zeroing; strata: number of interference-hit chirps among the 16 history chirps):

```

=== Part O: IPIX, 56 units, 74980 test segments ===
-- O1: look-0 Pfa (alpha 0.01) and texture-quintile Pfa/alpha (low -> high), spread --
IN1    Pfa 0.0099 | 1.16 1.05 1.12 0.93 0.66 | spread 1.76
OS1_8  Pfa 0.0108 | 1.48 1.19 1.12 0.90 0.72 | spread 2.06
OS1_12 Pfa 0.0102 | 1.25 1.10 1.09 0.87 0.79 | spread 1.59
IN4    Pfa 0.0098 | 1.07 1.01 0.92 1.00 0.91 | spread 1.17
OS4    Pfa 0.0100 | 1.28 1.02 0.97 0.90 0.85 | spread 1.50
NA4    Pfa 0.0095 | 0.72 0.94 0.94 0.95 1.20 | spread 1.67
CALoc  Pfa 0.0108 | 0.69 0.75 0.98 1.12 1.84 | spread 2.68
-- dwell Pfa at K=8 (design 0.01): INisum 0.0102 OS1_8sum 0.0108 PAMFH 0.0096 PAMFOS 0.0102 Calocsum 0.0105
-- O2: abrupt target (L=1, look 0) SCR for Pd 0.5; difference vs IN1 --
IN1    SCR -0.82 dB | minus IN1 0.00 [ 0.00, 0.00] LODD 0.00.. 0.00
OS1_8  SCR 1.17 dB | minus IN1 1.99 [ 0.89, 3.39] LODD 1.50.. 2.56
OS1_12 SCR -0.49 dB | minus IN1 0.33 [ 0.17, 0.64] LODD 0.21.. 0.42
IN4    SCR -1.25 dB | minus IN1 -0.43 [-0.60, -0.24] LODD -0.48..-0.37
OS4    SCR 0.01 dB | minus IN1 0.84 [ 0.20, 1.70] LODD 0.58.. 1.25
NA4    SCR -1.92 dB | minus IN1 -1.10 [-1.25, -0.95] LODD -1.14..-1.06
CALoc  SCR 9.04 dB | minus IN1 9.86 [ 6.39, 12.73] LODD 8.99..11.60
-- O3: persistent target (L=1), per-look Pd at SCR 10 dB, looks 0..7 --
IN1    0.883 0.711 0.637 0.477 0.086 0.012 0.004 0.002
OS1_8  0.801 0.754 0.744 0.730 0.714 0.698 0.677 0.650
OS1_12 0.871 0.811 0.796 0.774 0.740 0.056 0.006 0.003
IN4    0.891 0.725 0.563 0.257 0.021 0.003 0.001 0.001
OS4    0.844 0.825 0.804 0.748 0.699 0.669 0.624 0.009
NA4    0.914 0.883 0.702 0.573 0.433 0.290 0.205 0.183
CALoc  0.558 0.557 0.558 0.559 0.557 0.558 0.558 0.557
-- O4: ramps; first-look SCR for Pd .5 | SCR for Pd .5 within 8 looks | dwell SCR (K=8) --
pre L= 1: IN1 -0.8| -3.1 OS1_8 1.2| -1.2 OS1_12 -0.5| -2.8 OS4 0.0| -1.9 NA4 -1.9| -4.1 CALoc 9.0| 5.4
dwell Pd.5/.9: INisum -2.7/ 6.9 OS1_8sum -0.4/ 9.2 PAMFH -3.7/ 3.0 PAMFOS -2.9/ 4.5 Calocsum 8.0/ 16.4
pre L= 4: IN1 0.4| -2.5 OS1_8 1.8| -0.4 OS1_12 0.1| -2.2 OS4 0.5| -1.1 NA4 -1.7| -4.0 CALoc 9.0| 5.4
dwell Pd.5/.9: INisum -2.1/ 16.7 OS1_8sum 0.7/ 11.9 PAMFH -3.5/ 13.2 PAMFOS -2.6/ 12.2 Calocsum 8.0/ 16.4
pre L= 8: IN1 4.4| 0.3 OS1_8 2.9| 0.8 OS1_12 2.9| -0.1 OS4 2.7| 1.2 NA4 -0.9| -4.0 CALoc 9.1| 5.4
dwell Pd.5/.9: INisum 0.3/ 19.3 OS1_8sum 2.6/ 15.5 PAMFH -3.1/ 15.1 PAMFOS -2.0/ 19.5 Calocsum 8.0/ 16.3
pre L=16: IN1 nan| nan OS1_8 nan| nan OS1_12 nan| nan OS4 nan| nan NA4 18.7| 13.0 CALoc 9.1| 5.4
dwell Pd.5/.9: INisum nan/ nan OS1_8sum nan/ nan PAMFH -1.8/ 19.3 PAMFOS 3.1/ nan Calocsum 8.0/ 16.4
in L= 1: IN1 -0.8| -3.1 OS1_8 1.2| -1.2 OS1_12 -0.5| -2.8 OS4 0.0| -1.9 NA4 -1.9| -4.1 CALoc 9.0| 5.4
dwell Pd.5/.9: INisum -2.7/ 6.9 OS1_8sum -0.4/ 9.2 PAMFH -3.7/ 3.0 PAMFOS -2.9/ 4.5 Calocsum 8.0/ 16.4
in L= 4: IN1 11.0| -2.8 OS1_8 11.8| -0.6 OS1_12 11.4| -2.4 OS4 11.7| -1.3 NA4 8.9| -3.8 CALoc 21.0| 6.0
dwell Pd.5/.9: INisum -2.1/ 9.6 OS1_8sum 0.7/ 11.1 PAMFH -3.0/ 9.2 PAMFOS -2.2/ 10.6 Calocsum 9.3/ 17.7
in L= 8: IN1 16.2| -2.0 OS1_8 15.5| 0.7 OS1_12 16.0| -1.5 OS4 15.0| -0.1 NA4 14.3| -3.0 CALoc 24.8| 7.5
dwell Pd.5/.9: INisum -0.3/ 12.1 OS1_8sum 3.0/ 13.5 PAMFH -1.8/ 13.6 PAMFOS -0.8/ 15.0 Calocsum 12.0/ 20.4
in L=16: IN1 19.9| 3.1 OS1_8 20.8| 5.8 OS1_12 20.2| 3.8 OS4 19.8| 5.0 NA4 17.7| -0.2 CALoc nan| 13.6
dwell Pd.5/.9: INisum 5.6/ 18.2 OS1_8sum 8.3/ 18.1 PAMFH 2.8/ 19.6 PAMFOS 4.3/ 20.3 Calocsum 18.0/ 23.9
R-pre L=8 first-look SCR, IN1 minus OS1_8: 1.46 [-0.31, 2.95] LODD 0.80.. 1.99
dwell K=8 Pd=0.5: SCR(INisum) - SCR(PAMFH) = 1.01 [ 0.36, 1.67] LODD 0.70.. 1.18
dwell K=8 Pd=0.5: SCR(OS1_8sum) - SCR(PAMFH) = 3.31 [ 0.79, 6.23] LODD 1.95.. 4.20
dwell K=8 Pd=0.5: SCR(PAMFOS) - SCR(PAMFH) = 0.80 [ 0.09, 1.69] LODD 0.46.. 1.13
dwell K=8 Pd=0.5: SCR(OS1_8sum) - SCR(INisum) = 2.30 [ 0.43, 4.59] LODD 1.26.. 3.08
dwell K=8 Pd=0.9: SCR(INisum) - SCR(PAMFH) = 3.95 [ 3.32, 4.43] LODD 3.57.. 4.17
dwell K=8 Pd=0.9: SCR(OS1_8sum) - SCR(PAMFH) = 6.46 [ 5.34, 7.67] LODD 6.00.. 6.68
dwell K=8 Pd=0.9: SCR(PAMFOS) - SCR(PAMFH) = 1.49 [ 0.92, 2.20] LODD 1.27.. 1.77
dwell K=8 Pd=0.9: SCR(OS1_8sum) - SCR(INisum) = 2.52 [ 1.26, 4.13] LODD 2.04.. 3.09
-- O7: OS exact law vs measured coverage (MAE per unit) --
OS1_8 1-a=0.9: law 0.9052 measured 0.9008 MAE 0.0214
OS1_8 1-a=0.99: law 0.9921 measured 0.9892 MAE 0.0051
OS1_12 1-a=0.9: law 0.8972 measured 0.9009 MAE 0.0121
OS1_12 1-a=0.99: law 0.9899 measured 0.9898 MAE 0.0032
OS4 1-a=0.9: law 0.9050 measured 0.9009 MAE 0.0125
OS4 1-a=0.99: law 0.9924 measured 0.9900 MAE 0.0035

```

```

VERDICTS Part O
O1 spread 2.06 <= 2.0: FAILED
O2 OS1_8 - IN1 = 1.99 dB in [0, 1.5]: FAILED
O3 OS1_8 look-4 Pd 0.714 >= 0.5: OK
O4 R-pre L=8 IN1 - OS1_8 first-look SCR = 1.46 dB >= 3: FAILED
O5 fraction of INisum's Pd.9 gap to PAMF-H closed by OS1_8sum = -0.64 >= 0.5: FAILED
O6 PAMF-OS - PAMF-H at Pd .5 = 0.80 dB within 1: OK
O7 OS1_8 law MAE at 0.90 = 0.0214 <= 0.015: FAILED

```

```

=== Part J: JKU, 8 station x Rx units ===
-- run ref: chirp hit rate 0.000; zeroed fraction N/Z 0.0000/0.0492
[N] episodes non-hit Y by history hits {'O': 443520, '1-2': 0, '3-5': 0, '>=6': 0}; hit-Y 0
[N] IN1 Pfa by stratum 0.0101 nan nan nan | Y-hit nan | Pd010 0.499 nan nan nan | SCR50 10.0 nan nan nan
[N] OS1_8 Pfa by stratum 0.0102 nan nan nan | Y-hit nan | Pd010 0.452 nan nan nan | SCR50 10.8 nan nan nan
[N] CA16 Pfa by stratum 0.0103 nan nan nan | Y-hit nan | Pd010 0.550 nan nan nan | SCR50 9.3 nan nan nan
[N] OS16_8 Pfa by stratum 0.0103 nan nan nan | Y-hit nan | Pd010 0.496 nan nan nan | SCR50 10.1 nan nan nan
[Z] episodes non-hit Y by history hits {'O': 443520, '1-2': 0, '3-5': 0, '>=6': 0}; hit-Y 0
[Z] IN1 Pfa by stratum 0.0102 nan nan nan | Y-hit nan | Pd010 0.480 nan nan nan | SCR50 10.3 nan nan nan
[Z] OS1_8 Pfa by stratum 0.0101 nan nan nan | Y-hit nan | Pd010 0.435 nan nan nan | SCR50 11.0 nan nan nan
[Z] CA16 Pfa by stratum 0.0104 nan nan nan | Y-hit nan | Pd010 0.524 nan nan nan | SCR50 9.7 nan nan nan
[Z] OS16_8 Pfa by stratum 0.0106 nan nan nan | Y-hit nan | Pd010 0.472 nan nan nan | SCR50 10.4 nan nan nan

```

```
-- run A: chirp hit rate 0.464; zeroed fraction N/Z 0.0000/0.0761
[N] episodes non-hit Y by history hits {'0': 337680, '1-2': 140880, '3-5': 11520, '>=6': 227760}; hit-Y 626160
[N] IN1 Pfa by stratum 0.0098 0.0093 0.0070 0.0010 | Y-hit 0.0176 | Pd@10 0.350 0.287 0.271 0.112 | SCR50 12.4 13.4 13.7 17.7
[N] OS1_8 Pfa by stratum 0.0101 0.0103 0.0080 0.0020 | Y-hit 0.0179 | Pd@10 0.310 0.252 0.239 0.100 | SCR50 13.1 14.1 14.3 18.4
[N] CA16 Pfa by stratum 0.0107 0.0095 0.0071 0.0005 | Y-hit 0.0217 | Pd@10 0.388 0.334 0.314 0.131 | SCR50 11.7 12.5 12.8 16.3
[N] OS16_8 Pfa by stratum 0.0122 0.0112 0.0105 0.0010 | Y-hit 0.0305 | Pd@10 0.349 0.301 0.291 0.126 | SCR50 12.4 13.1 13.3 16.7
[Z] episodes non-hit Y by history hits {'0': 337680, '1-2': 140880, '3-5': 11520, '>=6': 227760}; hit-Y 626160
[Z] IN1 Pfa by stratum 0.0106 0.0101 0.0113 0.0103 | Y-hit 0.0107 | Pd@10 0.467 0.455 0.452 0.429 | SCR50 10.5 10.7 10.7 11.2
[Z] OS1_8 Pfa by stratum 0.0105 0.0097 0.0102 0.0102 | Y-hit 0.0100 | Pd@10 0.422 0.407 0.402 0.387 | SCR50 11.2 11.4 11.5 11.9
[Z] CA16 Pfa by stratum 0.0107 0.0102 0.0088 0.0110 | Y-hit 0.0104 | Pd@10 0.494 0.490 0.489 0.494 | SCR50 10.1 10.2 10.2 10.1
[Z] OS16_8 Pfa by stratum 0.0108 0.0103 0.0088 0.0112 | Y-hit 0.0108 | Pd@10 0.448 0.444 0.446 0.447 | SCR50 10.8 10.9 10.9 10.9
-- run C: chirp hit rate 0.052; zeroed fraction N/Z 0.0000/0.0712
[N] episodes non-hit Y by history hits {'0': 1023600, '1-2': 100080, '3-5': 112320, '>=6': 42960}; hit-Y 65040
[N] IN1 Pfa by stratum 0.0107 0.0132 0.0080 0.0043 | Y-hit 0.0288 | Pd@10 0.464 0.311 0.243 0.218 | SCR50 10.6 13.2 14.4 15.0
[N] OS1_8 Pfa by stratum 0.0108 0.0147 0.0084 0.0056 | Y-hit 0.0268 | Pd@10 0.419 0.285 0.221 0.187 | SCR50 11.3 13.8 15.1 15.8
[N] CA16 Pfa by stratum 0.0112 0.0123 0.0061 0.0033 | Y-hit 0.0434 | Pd@10 0.515 0.377 0.310 0.266 | SCR50 9.8 12.0 13.0 13.7
[N] OS16_8 Pfa by stratum 0.0111 0.0147 0.0068 0.0042 | Y-hit 0.0397 | Pd@10 0.466 0.345 0.275 0.233 | SCR50 10.6 12.5 13.7 14.4
[Z] episodes non-hit Y by history hits {'0': 1023600, '1-2': 100080, '3-5': 112320, '>=6': 42960}; hit-Y 65040
[Z] IN1 Pfa by stratum 0.0105 0.0100 0.0085 0.0113 | Y-hit 0.0106 | Pd@10 0.471 0.438 0.411 0.431 | SCR50 10.5 11.0 11.4 11.1
[Z] OS1_8 Pfa by stratum 0.0105 0.0103 0.0091 0.0119 | Y-hit 0.0103 | Pd@10 0.427 0.394 0.366 0.391 | SCR50 11.2 11.7 12.2 11.8
[Z] CA16 Pfa by stratum 0.0105 0.0102 0.0100 0.0123 | Y-hit 0.0105 | Pd@10 0.514 0.495 0.480 0.494 | SCR50 9.8 10.1 10.3 10.1
[Z] OS16_8 Pfa by stratum 0.0104 0.0102 0.0099 0.0126 | Y-hit 0.0120 | Pd@10 0.466 0.448 0.436 0.443 | SCR50 10.5 10.9 11.1 10.9

VERDICTS Part J
J1 run A >=6-hit stratum Pd@10 drop vs ref: IN1 0.387 (>=0.2?), OS1_8 0.353 (<= half?) -> FAILED [n>=6 stratum: 227760]
J2 run A: IN1 hit-Y Pfa N 0.0176 -> Z 0.0107; reduction 0.39 (>=0.5?) -> FAILED
J2 run C: IN1 hit-Y Pfa N 0.0288 -> Z 0.0106; reduction 0.63 (>=0.5?) -> OK
J3 run A overall SCR50: Z@OS1_8 - Z@IN1 = 0.71 [0.66,0.76] dB (<= +0.57) -> FAILED
overall SCR50 (non-hit Y, all strata): N-ref: IN1 10.0 OS1_8 10.8 CA16 9.3 OS16_8 10.1 | N-A: IN1 15.7 OS1_8 16.4 CA16 14.6 OS16_8 15.1 | N-C: IN1 11.4 OS1_8 12.1 CA16 10.5 OS16_8 11.3 | Z-ref: IN1 10.3 OS1_8 11.0 OS16_8 10.9
J4 ref: CA16 - IN1 SCR50 = -0.78 dB (>0?) -> FAILED
```

15 Audit of existing outcomes: conditional false-alarm rates, dependence, leave-one-day-out, fit diagnostics

Output of `r10/audit_existing.py` (descriptive and sensitivity analyses requested in review; no new runs):

```
=== 1. IPIX false-alarm rate by texture quintile (confirmatory, m = 16, +/-1024 proxy) ===
-- design alpha = 0.1: Pfa/alpha per quintile (low -> high texture), mean over units; exceedances pooled --
IN1 1.04 0.98 1.00 0.99 0.92 | spread 1.1x | exceedances 1557 1466 1501 1483 1373 of 74980 episodes
IN4 1.00 0.99 0.97 1.00 0.97 | spread 1.0x | exceedances 1495 1474 1456 1501 1455 of 74980 episodes
NA4 0.84 0.96 1.06 1.02 1.02 | spread 1.3x | exceedances 1252 1436 1591 1540 1538 of 74980 episodes
U 0.22 0.44 0.71 1.12 2.56 | spread 11.5x | exceedances 325 662 1057 1680 3846 of 74980 episodes
RMS 1.88 1.25 0.88 0.60 0.43 | spread 4.4x | exceedances 2813 1876 1321 899 636 of 74980 episodes
MLP 1.82 1.17 0.84 0.61 0.50 | spread 3.6x | exceedances 2732 1761 1264 927 749 of 74980 episodes
MON 0.79 0.86 0.98 1.05 1.13 | spread 1.4x | exceedances 1177 1277 1464 1579 1689 of 74980 episodes
-- design alpha = 0.01: Pfa/alpha per quintile (low -> high texture), mean over units; exceedances pooled --
IN1 1.18 1.03 1.14 0.91 0.67 | spread 1.7x | exceedances 175 155 172 137 101 of 74980 episodes
IN4 1.09 0.99 0.93 0.99 0.93 | spread 1.2x | exceedances 163 149 141 150 140 of 74980 episodes
NA4 0.72 0.94 0.94 0.93 1.22 | spread 1.7x | exceedances 107 139 142 139 184 of 74980 episodes
U 0.05 0.18 0.21 0.60 4.68 | spread 87.5x | exceedances 8 27 32 89 710 of 74980 episodes
RMS 2.62 1.25 0.83 0.43 0.27 | spread 9.8x | exceedances 390 187 124 64 40 of 74980 episodes
MLP 2.35 1.13 0.81 0.45 0.48 | spread 5.2x | exceedances 353 167 121 67 71 of 74980 episodes
MON 0.50 0.60 0.92 0.83 1.14 | spread 2.3x | exceedances 74 90 139 125 171 of 74980 episodes
-- maximum quintile deviation: observed vs binomial reference vs dependence-aware (time-shift) reference, alpha = 0.1 --
IN1 observed 0.036 | binomial ref 0.029 | time-shift ref 0.033
IN4 observed 0.032 | binomial ref 0.029 | time-shift ref 0.033
NA4 observed 0.043 | binomial ref 0.029 | time-shift ref 0.039
U observed 0.165 | binomial ref 0.029 | time-shift ref 0.062

=== 2. Dependence of IN1 exceedance indicators (alpha = 0.1), mean over units ===
lag-1 (256 pulses) autocorrelation within bin: -0.007 (range -0.067 to 0.046)
correlation across bins at the same time: 0.045

=== 3. Leave-one-day-out range of headline effects (estimate on all 6 days; min..max over the 6 LODD fits) ===
NA-AR(4)/IN-AR(4) radius ratio (0.90): 0.891 LODD 0.884..0.895
IN1 quintile Pfa spread (alpha 0.01): 1.75x LODD 1.72..1.83x
single-pulse gain IN1 vs CA16 Pd=0.5: 6.96 dB LODD 5.97.. 8.16
single-pulse gain IN1 vs Caloc Pd=0.5: 9.81 dB LODD 8.94..11.54
single-pulse gain NA4 vs IN1 Pd=0.5: 1.07 dB LODD 1.03.. 1.11
single-pulse gain NA4 vs IN4 Pd=0.5: 0.66 dB LODD 0.58.. 0.75
single-pulse gain IN1 vs CA16 Pd=0.9: 7.62 dB LODD 6.59.. 9.01
single-pulse gain IN1 vs Caloc Pd=0.9: 8.97 dB LODD 8.10..10.62
single-pulse gain NA4 vs IN1 Pd=0.9: 2.19 dB LODD 1.88.. 2.39
single-pulse gain NA4 vs IN4 Pd=0.9: 1.46 dB LODD 1.43.. 1.52
eight-look dwell gain IN1 vs Caloc-sum Pd=0.5: 10.70 dB LODD 9.86..12.38
eight-look dwell gain NA4 vs Caloc-sum Pd=0.5: 11.74 dB LODD 11.01..13.00
per-day values of the single-pulse IN1 vs Caloc gain (Pd 0.5): 1993-11-07 3.5, 1993-11-08 8.4, 1993-11-09 13.8, 1993-11-10 13.3, 1993-11-11 12.2, 1993-11-18 11.4

=== 4. Fit diagnostics (all confirmatory units, training thirds) ===
|r| of IN-ARCP: median 0.960, max 0.980; clamp |r| <= 0.98 binding in 5 of 56 units
NA-AR(2) Nelder-Mead converged in 56 of 56 units; NA-AR(4) in 16 of 56 (median iterations 1500, cap 1500)
whole-unit time (all 13 procedures, GPU): median 71 s

=== 5. Second radar: Pfa/alpha by CNR class (mean over units with data), exact law, counts ===
class (dB) <-5 -5..0 0..5 5..10 10..20 >=20
test episodes 721095 265416 229845 68556 31870 338
IN1 a=0.1 1.03 1.03 1.00 0.83 0.38 0.00 units: 24 24 24 24 24 2
NA4 a=0.1 0.83 1.36 1.19 0.88 0.54 0.00 units: 24 24 24 24 24 2
CA16 a=0.1 1.07 1.05 0.96 0.58 0.15 0.00 units: 24 24 24 24 24 2
law a=0.1 1.04 1.00 0.95 0.86 0.68 0.34 (exact law, 3-decimal coverage from jku_summary)
IN1 a=0.01 1.07 1.06 0.95 0.52 0.08 0.00 units: 24 24 24 24 24 2
NA4 a=0.01 0.69 1.68 1.27 0.66 0.22 0.00 units: 24 24 24 24 24 2
CA16 a=0.01 1.14 1.05 0.78 0.19 0.02 0.00 units: 24 24 24 24 24 2
law a=0.01 1.10 1.00 0.90 0.70 0.50 0.20 (exact law, 3-decimal coverage from jku_summary)
```

Exploratory: sensitivity of NA-AR(4) to optimizer convergence (restart from the capped solution for up to 4,500 further Nelder–Mead iterations; `r10/explore_na4_convergence.py`):

```
units 56; converged at cap 1500: 16; converged after restart (+4500): 55
log-likelihood gain per training episode: median 3.13e-06, max 1.54e-04
1-alpha=0.9: radius ratio long/orig geo-mean 0.9989 (range 0.9790..1.0132); coverage orig 0.9019 long 0.9020
1-alpha=0.99: radius ratio long/orig geo-mean 0.9997 (range 0.9492..1.0327); coverage orig 0.9905 long 0.9906
```

16 Proofs of the whitened-innovation laws

Throughout, $w \sim \mathcal{CN}(0, I)$ denotes whitened clutter innovations (unit variance after dividing by σ_e), $g \sim \mathcal{CN}(0, 1)$ the Swerling-1 amplitude, and u the whitened target signature, so that the whitened episode is $w + gu \sim \mathcal{CN}(0, I + uu^H)$. E , $E_{\pm}$ are standard exponentials and Γ_k a Gamma($k, 1$) variable, all independent.

16.1 Proposition (post-onset rank-one law)

Write the detection event as $Q > 0$ with $Q = x^H M x$, $x = w + gu$ and $M = \text{diag}(-\beta I_m, 1)$, $\beta = q^2/m$. As in Appendix A of the paper, $Q \stackrel{d}{=} \sum_k \lambda_k E_k$ with λ_k the eigenvalues of $C^{1/2} M C^{1/2}$, $C = I + uu^H$, which are those of $MC = M + (Mu)u^H$. For a rank-one update of a diagonal matrix,

$$\det(\lambda I - M - Muu^H) = \det(\lambda I - M) \left(1 - \sum_i \frac{M_{ii}|u_i|^2}{\lambda - M_{ii}}\right) = (\lambda - 1)(\lambda + \beta)^m \left(1 - \frac{a}{\lambda - 1} + \frac{\beta b}{\lambda + \beta}\right),$$

with $a = |u_Y|^2$ and $b = \|u_H\|^2$. Hence $-\beta$ is an eigenvalue of multiplicity $m - 1$ and the other two solve $(\lambda - 1)(\lambda + \beta) - a(\lambda + \beta) + \beta b(\lambda - 1) = 0$, which is the quadratic of the proposition. Their product is $-\beta(1 + a + b) < 0$, so $\lambda_+ > 0 > \lambda_-$. Then

$$\mathbb{P}\{Q > 0\} = \mathbb{P}\{\lambda_+ E_+ > |\lambda_-| E_- + \beta \Gamma_{m-1}\} = \mathbb{E} e^{-(|\lambda_-| E_- + \beta \Gamma_{m-1})/\lambda_+} = \frac{\lambda_+}{\lambda_+ + |\lambda_-|} \left(1 + \frac{\beta}{\lambda_+}\right)^{-(m-1)}.$$

For $b = 0$ the roots are $1 + a$ and $-\beta$, and the expression reduces to $(1 + \beta/(1 + a))^{-m}$. The signature of a persistent target follows from the whitened amplitudes in (4): the onset innovation carries S_w , every later one $S_w |d|^2$.

16.2 Corollary (visibility horizon)

With $a = S_w D$ and $b = S_w(1 + (\ell - 1)D)$, $D = |d(\omega)|^2$, the linear coefficient is $B = 1 - \beta + S_w \kappa$, $\kappa = D - \beta\{1 + (\ell - 1)D\}$, and $\lambda_{\pm} = \{B \pm (B^2 + 4\beta(1 + a + b))^{1/2}\}/2$ with $1 + a + b = O(S_w)$. If $\kappa > 0$, then $\lambda_+ \sim S_w \kappa \rightarrow \infty$ while λ_- stays bounded, so $P_d \rightarrow 1$. If $\kappa < 0$, then $\lambda_+ \rightarrow \beta(1 + \ell D)/|\kappa|$ stays bounded while $\lambda_- \sim S_w \kappa \rightarrow -\infty$, so $\lambda_+ / (\lambda_+ - \lambda_-) \rightarrow 0$ and $P_d \rightarrow 0$. The sign change $\kappa = 0$ is $\ell = 1 + 1/\beta - 1/D$.

16.3 Order-statistic scale

Conditional on $G = |g|^2$, the history powers $|w_j + gu_j|^2$ are independent, exponential for clean innovations and non-central (with $2|w_j + gu_j|^2 \sim \chi_2^2(2G|u_j|^2)$) for contaminated ones. The distribution function of their k -th smallest value $X_{(k)}$ is the Poisson-binomial probability that at least k are below x , and $P_d = \mathbb{E}_G \int \mathbb{P}\{|w_Y + gu_Y|^2 > q^2 x \mid G\} dF_{X_{(k)}|G}(x)$. The code evaluates this by midpoint quadrature in the distribution function of G and a dynamic program for the Poisson-binomial law. If $\ell \leq m - k$ contaminated powers diverge, $X_{(k)}$ is the k -th smallest of the $m - \ell$ clean ones, which is independent of w_Y , and the OS-CFAR law with $m - \ell$ cells gives (7). If $\ell > m - k$, $X_{(k)}$ diverges with the target while $|w_Y + gu_Y|^2 / X_{(k)}$ tends to a constant below any practical threshold, as in Proposition 3.

16.4 Dwell integration

Under H_0 the event is $\Gamma_K > \tau \Gamma_m$, and conditioning on Γ_m with the Poisson form of the Gamma tail gives $\sum_{i < K} \mathbb{E}[e^{-\tau \Gamma_m} (\tau \Gamma_m)^i / i!] = \sum_{i < K} \binom{m+i-1}{i} \tau^i (1 + \tau)^{-(m+i)}$. With a Swerling-1 target the dwell form has one eigenvalue $c = 1 + b_D$ and $K - 1$ unit ones. For $v > 0$,

$$\mathbb{P}\{cE + \Gamma_{K-1} > v\} = \mathbb{P}\{\Gamma_{K-1} > v\} + e^{-v/c} (1 - 1/c)^{-(K-1)} \mathbb{P}\{\Gamma_{K-1} \leq v(1 - 1/c)\}.$$

Averaging over $v = \tau \Gamma_m$ uses $\mathbb{E} e^{-s \Gamma_m} = (1 + s)^{-m}$ and $\mathbb{E}[e^{-s \Gamma_m} \mathbb{P}\{\Gamma_{K-1} > \kappa \Gamma_m \mid \Gamma_m\}] = \sum_{i \leq K-2} \binom{m+i-1}{i} \kappa^i (1 + s + \kappa)^{-(m+i)}$ with $s + \kappa = \tau$, which gives (9).

16.5 Self-normalized whitened Doppler

With a known on-grid Doppler, the whitened dwell vector splits into its DFT coefficient on the target bin, $|\cdot|^2 \sim (1 + NS_w |d|^2) E$, and the orthogonal remainder Γ_{N-1} . The statistic $X/(X + \Gamma_{N-1})$ exceeds t iff $X(1 - t) > t \Gamma_{N-1}$, which has probability $\{1 + t/((1 - t)(1 + NS_w |d|^2))\}^{-(N-1)}$; at $S_w = 0$ this is $(1 - t)^{N-1}$. For the maximum over the N bins, let I_j be independent exponentials with means μ_j and J a set of bins with $|J|t < 1$. Substituting $x_j = tT + y_j$ for $j \in J$, where $T = \sum_i I_i$, the map has Jacobian $1/(1 - |J|t)$ and turns the exponent into a sum of independent exponentials, so

$$\mathbb{P}\{I_j > tT \ \forall j \in J\} = \frac{1}{1 - |J|t} \prod_{i=1}^N \frac{1}{1 + \theta_J \mu_i}, \quad \theta_J = \frac{t \sum_{j \in J} \mu_j^{-1}}{1 - |J|t},$$

and zero if $|J|t \geq 1$. With all $\mu_i = 1$ this is $(1 - |J|t)^{N-1}$, and inclusion-exclusion gives Fisher's law $\sum_r (-1)^{r-1} \binom{N}{r} (1 - rt)^{N-1}$ (R. A. Fisher, Proc. R. Soc. Lond. A 125, 1929). With one bin of mean $1 + NS_w |d|^2$, inclusion-exclusion over sets that do or do not contain it gives P_d (`laws.pd_snd_max`).

16.6 Theorem (certified integration)

Domination. Let $\mathcal{H}_H$ and $\mathcal{H}_D$ be the corrupted history and dwell innovations. The k_H -th smallest of the history powers is a nondecreasing function of each power, so over all values of the corrupted entries it is smallest when they are zero; hence $s_{k_H}^2 \geq \underline{s}^2$, the value computed with those entries set to zero. Every uncorrupted dwell term is a clean power divided by $s_{k_H}^2 \geq \underline{s}^2$, and every clipped term is at most c . Therefore $T \leq W$ for all interference values. For binary integration, an uncorrupted exceedance of the actual statistic is an exceedance at $\underline{s}^2$, and a corrupted term counts at most one. Adding elements to $\mathcal{H}$ can only lower $\underline{s}^2$ and replace terms by their maximum, so W is nondecreasing in $\mathcal{H}$. *Calibration.* Let the test hit set $\mathcal{H}$ be independent of the clutter and let $\tilde{\mathcal{H}} \sim Q$ with $\mathcal{H} \subseteq \tilde{\mathcal{H}}$ (a coupling that exists under stochastic dominance). Then $T \leq W(X, \mathcal{H}) \leq W(X, \tilde{\mathcal{H}})$. The pairs $(X_i, \mathcal{H}_i)$ and $(X, \tilde{\mathcal{H}})$ are exchangeable, so the split-conformal rank argument gives $\mathbb{P}\{W(X, \tilde{\mathcal{H}}) > \hat{q}\} \leq \alpha$. No property of the interference values is used, so the bound holds for adversarial amplitudes chosen after seeing the clutter, as long as the hit positions are independent of the clutter. *Trimmed sums.* If a corrupted dwell term can take any value, the trimmed sum over the $K - t$ smallest terms is unbounded once more than t dwell terms are corrupted, which a Bernoulli hit model allows with positive probability. The conformal quantile of the worst case is then infinite whenever that probability exceeds α .

16.7 Conformal calibration size and loss

For continuous i.i.d. scores the k -th order statistic of n calibration scores, mapped through the score distribution function, is $\text{Beta}(k, n + 1 - k)$; the conditional false-alarm probability is one minus it, hence $\text{Beta}(n + 1 - k, k)$. The smallest n with $\mathbb{P}\{P_{\text{fa}} > 2\alpha\} \leq 0.05$ is found by search (`laws.n_required`). For $P_d = P_{\text{fa}}^\gamma$, $\mathbb{E}U^\gamma = B(n + 1 - k + \gamma, k)/B(n + 1 - k, k)$.

17 R12: Monte Carlo checks, full results and verdicts

Frozen protocol `r12/PROTOCOL_R12.md` (sha256 a32706e7...); expectations and verdicts in `r12/RESULTS_R12.md` (12 of 18 met). Monte Carlo checks of every closed form in Section V (`r12/verify_laws.py`, 400,000 replications):

```
V1-V3 IN-ARCP per-look score: onset, post-onset rank-one law, visibility horizon
Pd clutter Doppler, SCR 10 dB, look 0 theory 0.93148 MC 0.93112 z -0.46
Pd clutter Doppler, SCR 10 dB, look 1 theory 0.00057 MC 0.00062 z +0.71
Pd clutter Doppler, SCR 10 dB, look 2 theory 0.00056 MC 0.00057 z +0.08
Pd clutter Doppler, SCR 10 dB, look 4 theory 0.00056 MC 0.00051 z -0.65
Pd clutter Doppler, SCR 10 dB, look 8 theory 0.00055 MC 0.00062 z +0.98
visibility horizon (clutter): l* = -200.08
Pd clutter Doppler, SCR 25 dB, look 0 theory 0.99772 MC 0.99770 z -0.16
Pd clutter Doppler, SCR 25 dB, look 1 theory 0.00002 MC 0.00001 z -0.65
Pd clutter Doppler, SCR 25 dB, look 2 theory 0.00002 MC 0.00002 z +0.08
Pd clutter Doppler, SCR 25 dB, look 4 theory 0.00002 MC 0.00002 z +0.10
Pd clutter Doppler, SCR 25 dB, look 8 theory 0.00002 MC 0.00002 z +0.13
visibility horizon (clutter): l* = -200.08
Pd opposite Doppler, SCR 10 dB, look 0 theory 0.93148 MC 0.93064 z -1.06
Pd opposite Doppler, SCR 10 dB, look 1 theory 0.97857 MC 0.97844 z -0.28
Pd opposite Doppler, SCR 10 dB, look 2 theory 0.96169 MC 0.96205 z +0.59
Pd opposite Doppler, SCR 10 dB, look 4 theory 0.15389 MC 0.15473 z +0.74
Pd opposite Doppler, SCR 10 dB, look 8 theory 0.00045 MC 0.00043 z -0.25
visibility horizon (opposite): l* = 3.73
Pd opposite Doppler, SCR 25 dB, look 0 theory 0.99772 MC 0.99778 z +0.37
Pd opposite Doppler, SCR 25 dB, look 1 theory 0.99931 MC 0.99939 z +0.95
Pd opposite Doppler, SCR 25 dB, look 2 theory 0.99875 MC 0.99890 z +1.37
Pd opposite Doppler, SCR 25 dB, look 4 theory 0.01381 MC 0.01368 z -0.36
Pd opposite Doppler, SCR 25 dB, look 8 theory 0.00001 MC 0.00001 z -0.36
visibility horizon (opposite): l* = 3.73
random Doppler (closed form averaged over w) at 10 dB:
Pd random Doppler, 10 dB, look 0 theory 0.93148 MC 0.93113 z -0.63
Pd random Doppler, 10 dB, look 1 theory 0.74117 MC 0.74048 z -0.69
Pd random Doppler, 10 dB, look 2 theory 0.67605 MC 0.67458 z -1.41
Pd random Doppler, 10 dB, look 4 theory 0.08049 MC 0.08077 z +0.45
Pd random Doppler, 10 dB, look 8 theory 0.00080 MC 0.00081 z +0.31

V4 order-statistic score (k = 8, m = 16): Pfa law and post-onset law
Pfa OS (k=8) theory 0.01000 MC 0.01003 z +0.17
Pd OS exact, quarter Doppler, 10 dB, look 0 theory 0.92017 MC 0.92008 z -0.10
Pd OS exact, quarter Doppler, 10 dB, look 2 theory 0.94773 MC 0.94665 z -1.53
strong-target limit: 0.9471
Pd OS exact, quarter Doppler, 10 dB, look 6 theory 0.90930 MC 0.90896 z -0.38
strong-target limit: 0.9079
Pd OS exact, quarter Doppler, 10 dB, look 8 theory 0.83542 MC 0.83647 z +0.89
strong-target limit: 0.8339
Pd OS exact, quarter Doppler, 10 dB, look 9 theory 0.00136 MC 0.00107 z -2.48
strong-target limit: 0.0000
Pd OS exact, quarter Doppler, 25 dB, look 0 theory 0.99735 MC 0.99717 z -1.11
Pd OS exact, quarter Doppler, 25 dB, look 2 theory 0.99834 MC 0.99796 z -2.91
strong-target limit: 0.9983
Pd OS exact, quarter Doppler, 25 dB, look 6 theory 0.99695 MC 0.99662 z -1.88
strong-target limit: 0.9969
Pd OS exact, quarter Doppler, 25 dB, look 8 theory 0.99418 MC 0.99366 z -2.16
strong-target limit: 0.9941
Pd OS exact, quarter Doppler, 25 dB, look 9 theory 0.00004 MC 0.00005 z +0.25
strong-target limit: 0.0000

V5 dwell integration with a shared history (m = 16): Pfa and Pd laws
Pfa dwell K=2 theory 0.01000 MC 0.00996 z -0.16
Pd dwell K=2, 0 dB, quarter Doppler theory 0.73521 MC 0.73494 z -0.27
```

```

Pd dwell K=2, 10 dB, quarter Doppler      theory 0.96800 MC 0.96865 z +1.65
Pfa dwell K=4                             theory 0.01000 MC 0.00982 z -0.81
Pd dwell K=4, 0 dB, quarter Doppler       theory 0.82829 MC 0.82673 z -1.84
Pd dwell K=4, 10 dB, quarter Doppler      theory 0.98077 MC 0.98072 z -0.14
Pfa dwell K=8                             theory 0.01000 MC 0.01006 z +0.27
Pd dwell K=8, 0 dB, quarter Doppler       theory 0.87697 MC 0.87728 z +0.42
Pd dwell K=8, 10 dB, quarter Doppler      theory 0.98671 MC 0.98684 z +0.51
power integration over 8 pulses at |rho| = 0.93: effective looks K_eff = 1.41

V6 self-normalized whitened Doppler (P-ANMF): Fisher's g and the NMF law
Pfa known bin K=8                        theory 0.01000 MC 0.00995 z -0.16
Pfa max (Fisher g) K=8                  theory 0.01000 MC 0.01001 z +0.03
Pd known bin K=8, -10 dB                 theory 0.74050 MC 0.74046 z -0.03
Pd max K=8, -10 dB                       theory 0.60215 MC 0.60083 z -0.85
Pd known bin K=8, 0 dB                   theory 0.96852 MC 0.96817 z -0.64
Pd max K=8, 0 dB                         theory 0.94670 MC 0.94600 z -0.99
Pfa known bin K=16                       theory 0.01000 MC 0.00967 z -1.05
Pfa max (Fisher g) K=16                  theory 0.01000 MC 0.00983 z -0.54
Pd known bin K=16, -10 dB                theory 0.75351 MC 0.75314 z -0.27
Pd max K=16, -10 dB                      theory 0.60966 MC 0.61132 z +1.07
Pd known bin K=16, 0 dB                  theory 0.97047 MC 0.97081 z +0.63
Pd max K=16, 0 dB                        theory 0.94856 MC 0.94997 z +2.01
Pfa known bin K=64                       theory 0.01000 MC 0.01024 z +0.38
Pfa max (Fisher g) K=64                  theory 0.01000 MC 0.01056 z +0.89
Pd known bin K=64, -10 dB                theory 0.66765 MC 0.66292 z -1.59
Pd max K=64, -10 dB                      theory 0.45636 MC 0.45604 z -0.10
Pd known bin K=64, 0 dB                  theory 0.95709 MC 0.95716 z +0.06
Pd max K=64, 0 dB                        theory 0.91784 MC 0.91952 z +0.97

V7 ANMF with the sample covariance (Kraut-Scharf law), Gaussian and compound-Gaussian data
Pfa ANMF-SCM N=8, Ks=24, Gaussian        theory 0.01000 MC 0.00990 z -0.14
Pfa ANMF-FP N=8, Ks=24, Gaussian [asymptotic law] theory 0.01000 MC 0.00890 z -1.56
Pfa ANMF-SCM N=8, Ks=24, CG secondary+test [law assumes Gaussian] theory 0.01000 MC 0.03435 z +34.61
Pfa ANMF-FP N=8, Ks=24, CG secondary+test [asymptotic law] theory 0.01000 MC 0.00920 z -1.14
Pfa ANMF-SCM N=8, Ks=40, Gaussian        theory 0.01000 MC 0.00980 z -0.28
Pfa ANMF-FP N=8, Ks=40, Gaussian [asymptotic law] theory 0.01000 MC 0.00955 z -0.64
Pfa ANMF-SCM N=8, Ks=40, CG secondary+test [law assumes Gaussian] theory 0.01000 MC 0.02795 z +25.51
Pfa ANMF-FP N=8, Ks=40, CG secondary+test [asymptotic law] theory 0.01000 MC 0.01105 z +1.49

V8 conformal calibration: conditional Pfa ~ Beta(n+1-k, k); expected Pd
n=1000, alpha=0.01: mean Pfa theory 0.00999 MC 0.01000; P(Pfa > 2 alpha) theory 0.0047 MC 0.0049
E[Pd] at SNR 10 dB (Pd = Pfa^(1/(1+SNR))): theory 0.65515 MC 0.65521
n=2000, alpha=0.001: mean Pfa theory 0.00100 MC 0.00100; P(Pfa > 2 alpha) theory 0.0914 MC 0.0909
E[Pd] at SNR 10 dB (Pd = Pfa^(1/(1+SNR))): theory 0.52206 MC 0.52211
calibration size for P(Pfa > 2 alpha) <= 0.05 at alpha = 0.01: n >= 153
calibration size for P(Pfa > 2 alpha) <= 0.05 at alpha = 0.001: n >= 1559
calibration size for P(Pfa > 2 alpha) <= 0.05 at alpha = 0.0001: n >= 15520

V9 certified integration under pulsed interference (m = 16, K = 8, median history, AR(1))
q: conformal quantile of the worst case over amplitudes, on clean episodes with hit positions drawn at
the design rate p. Test hits at rate p with Gaussian amplitudes (JNR re local power) or exact cancellation.
clip p=0.0 JNR 30 dB : violations 0e+00; Pfa certified 0.0106, clean-thr 0.0106, CA-NCI 0.0101; Pd(10 dB) certified 0.978, clean-thr 0.978
clip p=0.02 JNR 30 dB : violations 0e+00; Pfa certified 0.0034, clean-thr 0.0169, CA-NCI 0.1324; Pd(10 dB) certified 0.967, clean-thr 0.977
clip p=0.02 JNR 10 dB : violations 0e+00; Pfa certified 0.0032, clean-thr 0.0167, CA-NCI 0.1170; Pd(10 dB) certified 0.967, clean-thr 0.977
clip p=0.02 cancelling : violations 0e+00; Pfa certified 0.0029, clean-thr 0.0144, CA-NCI 0.0528; Pd(10 dB) certified 0.967, clean-thr 0.977
clip p=0.05 JNR 30 dB : violations 0e+00; Pfa certified 0.0003, clean-thr 0.0316, CA-NCI 0.2266; Pd(10 dB) certified 0.938, clean-thr 0.975
clip p=0.05 JNR 10 dB : violations 0e+00; Pfa certified 0.0003, clean-thr 0.0297, CA-NCI 0.2016; Pd(10 dB) certified 0.933, clean-thr 0.975
clip p=0.05 cancelling : violations 0e+00; Pfa certified 0.0002, clean-thr 0.0218, CA-NCI 0.0884; Pd(10 dB) certified 0.936, clean-thr 0.976
trim p=0.0 JNR 30 dB : violations 0e+00; Pfa certified 0.0103, clean-thr 0.0103, CA-NCI 0.0096; Pd(10 dB) certified 0.978, clean-thr 0.978
trim p=0.02 JNR 30 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0286, CA-NCI 0.1332; Pd(10 dB) certified 0.000, clean-thr 0.977
trim p=0.02 JNR 10 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0242, CA-NCI 0.1175; Pd(10 dB) certified 0.000, clean-thr 0.976
trim p=0.02 cancelling : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0166, CA-NCI 0.0529; Pd(10 dB) certified 0.000, clean-thr 0.976
trim p=0.05 JNR 30 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0821, CA-NCI 0.2260; Pd(10 dB) certified 0.000, clean-thr 0.976
trim p=0.05 JNR 10 dB : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0659, CA-NCI 0.2016; Pd(10 dB) certified 0.000, clean-thr 0.975
trim p=0.05 cancelling : violations 0e+00; Pfa certified 0.0000, clean-thr 0.0297, CA-NCI 0.0880; Pd(10 dB) certified 0.000, clean-thr 0.975
bin p=0.0 JNR 30 dB : violations 0e+00; Pfa certified 0.0043, clean-thr 0.0043, CA-NCI 0.0100; Pd(10 dB) certified 0.967, clean-thr 0.967
bin p=0.02 JNR 30 dB : violations 0e+00; Pfa certified 0.0028, clean-thr 0.0135, CA-NCI 0.1313; Pd(10 dB) certified 0.959, clean-thr 0.968
bin p=0.02 JNR 10 dB : violations 0e+00; Pfa certified 0.0026, clean-thr 0.0127, CA-NCI 0.1174; Pd(10 dB) certified 0.958, clean-thr 0.967
bin p=0.02 cancelling : violations 0e+00; Pfa certified 0.0020, clean-thr 0.0090, CA-NCI 0.0522; Pd(10 dB) certified 0.958, clean-thr 0.967
bin p=0.05 JNR 30 dB : violations 0e+00; Pfa certified 0.0003, clean-thr 0.0464, CA-NCI 0.2271; Pd(10 dB) certified 0.928, clean-thr 0.965
bin p=0.05 JNR 10 dB : violations 0e+00; Pfa certified 0.0002, clean-thr 0.0396, CA-NCI 0.2045; Pd(10 dB) certified 0.927, clean-thr 0.965
bin p=0.05 cancelling : violations 0e+00; Pfa certified 0.0001, clean-thr 0.0200, CA-NCI 0.0891; Pd(10 dB) certified 0.929, clean-thr 0.965

Summary: 65 probability checks, max |z| = 2.91, 0 with |z| > 3; runtime 274 s

```

IPIX parts L (low false-alarm calibration), A (dwell detection with ANMF baselines), T (real target) and I (injected pulsed interference), with 95% day-cluster intervals (r12/analyze_r12.py):

R12 IPIX analysis: 56 units

```

Part L: pooled Pfa / alpha [95% day-cluster interval]; texture-quintile spread (max/min)
alpha = 0.01
ANMF-FP(w) conf 1.04 [0.98, 1.07] spread 2.6
ANMF-FP(w) anal 1.62 [1.39, 1.82] spread 2.5
ANMF-SCM(w) conf 1.06 [0.99, 1.12] spread 2.3
ANMF-SCM(w) anal 2.02 [1.69, 2.28] spread 2.2
CA16 conf 1.00 [0.99, 1.01] spread 1.1
CA16 anal 0.80 [0.71, 0.85] spread 1.0
Clip-OS1 conf 1.05 [0.99, 1.08] spread 2.0
Clip-OS1 anal 4.84 [3.72, 6.72] spread 1.2
D-IN4 conf 1.02 [0.98, 1.07] spread 1.7
D-IN4 anal 2.02 [1.73, 2.27] spread 1.6
IN1 conf 1.01 [0.99, 1.02] spread 1.3
IN1 anal 0.96 [0.92, 1.02] spread 1.3
IN1 boot 1.63 [1.47, 1.75] spread 1.3
IN4 conf 1.00 [0.99, 1.01] spread 1.1
IN4 anal 1.03 [1.01, 1.05] spread 1.1
NA4 conf 1.00 [0.95, 1.08] spread 1.9
NA4 anal 1.56 [1.40, 1.70] spread 1.6
OS1_8 conf 1.04 [1.01, 1.08] spread 1.7
OS1_8 anal 2.15 [1.06, 3.57] spread 2.1
OSraw conf 1.01 [0.96, 1.04] spread 1.1
OSraw anal 2.07 [1.47, 2.55] spread 1.1
P-ANMF conf 1.04 [0.95, 1.11] spread 4.4
P-ANMF anal 4.36 [3.58, 5.29] spread 2.3
P-ANMF(bin) conf 1.10 [1.01, 1.19] spread 1.5
P-ANMF(bin) anal 1.78 [0.73, 2.61] spread 2.3
PAMF-H(w) conf 0.99 [0.96, 1.03] spread 2.2
PAMF-H(w) anal 1.97 [1.06, 2.70] spread 2.2
alpha = 0.001
ANMF-FP(w) conf 1.05 [0.96, 1.12] spread 2.1
ANMF-FP(w) anal 2.66 [1.76, 3.48] spread 2.2
ANMF-SCM(w) conf 1.04 [0.94, 1.14] spread 1.8
ANMF-SCM(w) anal 3.57 [2.38, 4.54] spread 2.0
CA16 conf 1.00 [0.94, 1.06] spread 1.1

```

```

CA16      anal      0.56 [0.41, 0.63] spread 1.3
Clip-OS1   conf      0.67 [0.34, 0.95] spread 12.1
Clip-OS1   anal     17.19 [11.68, 25.51] spread 1.3
D-IN4      conf      1.05 [0.86, 1.20] spread 1.8
D-IN4      anal      4.35 [3.16, 5.33] spread 2.1
IN1        conf      1.03 [1.01, 1.04] spread 1.5
IN1        anal      1.02 [0.91, 1.12] spread 1.4
IN1        boot      2.43 [2.15, 2.67] spread 1.4
IN4        conf      1.02 [0.96, 1.08] spread 1.1
IN4        anal      1.19 [1.01, 1.31] spread 1.1
NA4        conf      1.05 [1.00, 1.08] spread 3.0
NA4        anal      2.79 [2.38, 3.24] spread 2.3
OS1_8      conf      1.02 [0.89, 1.13] spread 1.9
OS1_8      anal      8.34 [2.00, 15.87] spread 4.0
OSraw      conf      0.98 [0.91, 1.02] spread 1.9
OSraw      anal      5.56 [3.15, 7.36] spread 1.4
P-ANMF     conf      1.04 [0.90, 1.16] spread 9.9
P-ANMF     anal     13.06 [10.39, 17.19] spread 4.5
P-ANMF(bin) conf      1.14 [0.95, 1.26] spread 1.7
P-ANMF(bin) anal      3.05 [1.01, 5.06] spread 2.4
PAMF-H(w)  conf      1.00 [0.89, 1.08] spread 2.1
PAMF-H(w)  anal      3.66 [1.66, 5.60] spread 2.4

alpha = 0.0001
ANMF-FP(w) conf      0.82 [0.67, 0.95] spread 2.0
ANMF-FP(w) anal      6.01 [2.87, 9.15] spread 1.8
ANMF-SCM(w) conf      0.85 [0.75, 1.01] spread 3.2
ANMF-SCM(w) anal      8.07 [3.98, 12.11] spread 1.8
CA16      conf      0.75 [0.68, 0.90] spread 1.3
CA16      anal      0.49 [0.26, 0.70] spread 1.6
Clip-OS1   conf      0.03 [0.00, 0.11] spread 6272109.2
Clip-OS1   anal     59.89 [40.97, 85.35] spread 1.5
D-IN4      conf      1.02 [0.59, 1.31] spread 2.2
D-IN4      anal     14.25 [7.84, 19.14] spread 2.0
IN1        conf      0.73 [0.54, 0.99] spread 2.7
IN1        anal      1.92 [1.26, 2.61] spread 2.0
IN1        boot      4.27 [3.44, 4.74] spread 1.3
IN4        conf      0.86 [0.64, 1.11] spread 1.9
IN4        anal      2.73 [1.48, 3.67] spread 1.3
NA4        conf      1.04 [0.84, 1.19] spread 2.9
NA4        anal      7.46 [4.55, 9.73] spread 3.2
OS1_8      conf      0.92 [0.76, 1.14] spread 3.0
OS1_8      anal     45.91 [7.96, 90.46] spread 7.1
OSraw      conf      0.75 [0.57, 1.07] spread 3.3
OSraw      anal     18.33 [8.92, 25.59] spread 1.8
P-ANMF     conf      1.47 [0.75, 2.57] spread 23.2
P-ANMF     anal     49.61 [35.38, 73.34] spread 7.0
P-ANMF(bin) conf      1.10 [0.79, 1.31] spread 2.0
P-ANMF(bin) anal      5.47 [1.35, 9.99] spread 2.0
PAMF-H(w)  conf      0.89 [0.55, 1.22] spread 4.0
PAMF-H(w)  anal      9.88 [4.13, 15.83] spread 2.4

L1 conformal within band: False; offenders: [['Clip-OS1', 0.001, np.float64(0.67)], ['Clip-OS1', 0.0001, np.float64(0.03)]]
L2 analytic whitened rules > 2x at 1e-4: ['IN4', 'OS1_8', 'NA4', 'D-IN4', 'PAMF-H(w)', 'P-ANMF(bin)', 'P-ANMF', 'Clip-OS1']
L3 False; L4 False; L5 True; L6 True

Part A: SCR (dB) for Pd = 0.5 / 0.9 from pooled curves (alpha = 0.01 and 0.001), measured Pfa
0.01 ANMF-FP SCR50 0.36 SCR90 20.26 Pfa 0.0101
0.01 ANMF-SCM SCR50 -0.18 SCR90 20.27 Pfa 0.0099
0.01 Bin-OS1 SCR50 -1.20 SCR90 14.49 Pfa 0.0065
0.01 Clip-OS1 SCR50 -2.32 SCR90 12.99 Pfa 0.0102
0.01 D-IN1 SCR50 -4.36 SCR90 8.34 Pfa 0.0103
0.01 D-IN4 SCR50 -5.00 SCR90 7.05 Pfa 0.0097
0.01 Exc-D-IN1 SCR50 nan SCR90 nan Pfa 0.0099
0.01 IN1look SCR50 -1.22 SCR90 10.98 Pfa 0.0099
0.01 OSraw SCR50 9.41 SCR90 19.46 Pfa 0.0108
0.01 P-ANMF SCR50 0.20 SCR90 nan Pfa 0.0102
0.01 PAMF-H SCR50 -5.00 SCR90 5.13 Pfa 0.0096
0.001 ANMF-FP SCR50 9.77 SCR90 nan Pfa 0.0010
0.001 ANMF-SCM SCR50 9.53 SCR90 nan Pfa 0.0009
0.001 Bin-OS1 SCR50 nan SCR90 nan Pfa 0.0002
0.001 Clip-OS1 SCR50 nan SCR90 nan Pfa 0.0004
0.001 D-IN1 SCR50 -0.14 SCR90 12.44 Pfa 0.0009
0.001 D-IN4 SCR50 -1.65 SCR90 12.52 Pfa 0.0007
0.001 Exc-D-IN1 SCR50 nan SCR90 nan Pfa 0.0007
0.001 IN1look SCR50 1.86 SCR90 14.12 Pfa 0.0011
0.001 OSraw SCR50 14.35 SCR90 24.54 Pfa 0.0007
0.001 P-ANMF SCR50 nan SCR90 nan Pfa 0.0011
0.001 PAMF-H SCR50 -4.25 SCR90 9.88 Pfa 0.0010
A1 ANMF-FP minus PAMF-H: 5.36 dB [1.93, 8.83] (expect >= 1)
A2 Clip-OS1 minus D-IN1: 2.04 dB [0.00, 2.90] (expect <= 1.5)
A3 OSraw minus IN1 (first look): 10.62 dB [7.05, 12.69] (expect >= 5)

Part T: real target cell. Mean Pd over units (clutter Pfa), alpha = 1e-3 [1e-2]
P-ANMF conf N=8 Pd 0.217 (Pfa 0.0011) [Pd 0.352]
P-ANMF conf N=16 Pd 0.224 (Pfa 0.0013) [Pd 0.362]
P-ANMF conf N=32 Pd 0.225 (Pfa 0.0013) [Pd 0.388]
P-ANMF conf N=64 Pd 0.297 (Pfa 0.0013) [Pd 0.469]
P-ANMF conf N=128 Pd 0.372 (Pfa 0.0012) [Pd 0.538]
P-ANMF conf N=256 Pd 0.399 (Pfa 0.0015) [Pd 0.573]
P-ANMF conf N=512 Pd 0.367 (Pfa 0.0020) [Pd 0.567]
P-ANMF conf N=1024 Pd 0.409 (Pfa 0.0026) [Pd 0.581]
P-ANMF anal N=8 Pd 0.366 (Pfa 0.0127) [Pd 0.443]
P-ANMF anal N=16 Pd 0.447 (Pfa 0.0319) [Pd 0.528]
P-ANMF anal N=32 Pd 0.563 (Pfa 0.0689) [Pd 0.629]
P-ANMF anal N=64 Pd 0.669 (Pfa 0.1206) [Pd 0.724]
P-ANMF anal N=128 Pd 0.747 (Pfa 0.1781) [Pd 0.793]
P-ANMF anal N=256 Pd 0.794 (Pfa 0.2352) [Pd 0.837]
P-ANMF anal N=512 Pd 0.829 (Pfa 0.2986) [Pd 0.868]
P-ANMF anal N=1024 Pd 0.868 (Pfa 0.4038) [Pd 0.905]
CA-NCI conf N=8 Pd 0.003 (Pfa 0.0011) [Pd 0.014]
CA-NCI conf N=16 Pd 0.003 (Pfa 0.0011) [Pd 0.015]
CA-NCI conf N=32 Pd 0.004 (Pfa 0.0012) [Pd 0.015]
CA-NCI conf N=64 Pd 0.004 (Pfa 0.0013) [Pd 0.016]
CA-NCI conf N=128 Pd 0.004 (Pfa 0.0014) [Pd 0.016]
CA-NCI conf N=256 Pd 0.004 (Pfa 0.0017) [Pd 0.016]
CA-NCI conf N=512 Pd 0.005 (Pfa 0.0018) [Pd 0.016]
CA-NCI conf N=1024 Pd 0.007 (Pfa 0.0023) [Pd 0.014]
ANMF-FP conf N=8 Pd 0.129 (Pfa 0.0011) [Pd 0.291]
ANMF-FP conf N=16 Pd 0.179 (Pfa 0.0012) [Pd 0.328]
history-normalized on target: IN1 Pd 0.0004 (clutter Pfa 0.0010); Clip-OS1 Pd 0.0004 (clutter Pfa 0.0007)
T1 True; T2 True (0.86 of units); T3 True (1.00 of units)
T4 descriptive: P-ANMF N=1024 mean Pd at 1e-3: 0.409

Part I: injected pulsed interference. Pooled Pfa / alpha [upper day-cluster bound]; SCR50
p=0.02 JNR=10 D-IN1 Pfa/a 10.27 (upper 11.45) SCR50 -2.18
p=0.02 JNR=10 PAMF-H Pfa/a 10.76 (upper 12.67) SCR50 -5.00
p=0.02 JNR=10 P-ANMF Pfa/a 0.82 (upper 0.89) SCR50 3.03
p=0.02 JNR=10 ANMF-FP Pfa/a 1.00 (upper 1.18) SCR50 2.72

```

p=0.02 JNR=10 D-IN4	Pfa/a	11.21	(upper 12.68)	SCR50	-4.45
p=0.02 JNR=10 Clip-OS1	Pfa/a	1.04	(upper 1.12)	SCR50	-2.19
p=0.02 JNR=10 Clip-OS1-cert	Pfa/a	0.52	(upper 0.60)	SCR50	-0.85
p=0.02 JNR=10 Bin-OS1	Pfa/a	0.73	(upper 0.85)	SCR50	-1.08
p=0.02 JNR=10 Bin-OS1-cert	Pfa/a	0.29	(upper 0.34)	SCR50	0.17
p=0.02 JNR=10 Exc-D-IN1	Pfa/a	1.22	(upper 1.29)	SCR50	nan
p=0.02 JNR=30 D-IN1	Pfa/a	12.89	(upper 13.04)	SCR50	-1.45
p=0.02 JNR=30 PAMF-H	Pfa/a	13.70	(upper 14.14)	SCR50	-5.00
p=0.02 JNR=30 P-ANMF	Pfa/a	0.82	(upper 0.89)	SCR50	5.75
p=0.02 JNR=30 ANMF-FP	Pfa/a	1.07	(upper 1.22)	SCR50	3.43
p=0.02 JNR=30 D-IN4	Pfa/a	13.53	(upper 13.91)	SCR50	-3.85
p=0.02 JNR=30 Clip-OS1	Pfa/a	1.07	(upper 1.15)	SCR50	-2.24
p=0.02 JNR=30 Clip-OS1-cert	Pfa/a	0.55	(upper 0.61)	SCR50	-0.89
p=0.02 JNR=30 Bin-OS1	Pfa/a	0.75	(upper 0.88)	SCR50	-1.10
p=0.02 JNR=30 Bin-OS1-cert	Pfa/a	0.32	(upper 0.36)	SCR50	0.15
p=0.02 JNR=30 Exc-D-IN1	Pfa/a	1.09	(upper 1.17)	SCR50	nan
p=0.05 JNR=10 D-IN1	Pfa/a	17.09	(upper 18.76)	SCR50	1.11
p=0.05 JNR=10 PAMF-H	Pfa/a	17.65	(upper 20.62)	SCR50	-2.39
p=0.05 JNR=10 P-ANMF	Pfa/a	0.58	(upper 0.69)	SCR50	6.14
p=0.05 JNR=10 ANMF-FP	Pfa/a	1.01	(upper 1.18)	SCR50	5.57
p=0.05 JNR=10 D-IN4	Pfa/a	19.12	(upper 21.44)	SCR50	-0.70
p=0.05 JNR=10 Clip-OS1	Pfa/a	1.17	(upper 1.28)	SCR50	-1.98
p=0.05 JNR=10 Clip-OS1-cert	Pfa/a	0.10	(upper 0.11)	SCR50	7.37
p=0.05 JNR=10 Bin-OS1	Pfa/a	1.05	(upper 1.37)	SCR50	-0.87
p=0.05 JNR=10 Bin-OS1-cert	Pfa/a	0.05	(upper 0.08)	SCR50	8.42
p=0.05 JNR=10 Exc-D-IN1	Pfa/a	1.55	(upper 1.83)	SCR50	nan
p=0.05 JNR=30 D-IN1	Pfa/a	21.23	(upper 21.72)	SCR50	10.96
p=0.05 JNR=30 PAMF-H	Pfa/a	22.12	(upper 23.12)	SCR50	4.33
p=0.05 JNR=30 P-ANMF	Pfa/a	0.55	(upper 0.60)	SCR50	18.56
p=0.05 JNR=30 ANMF-FP	Pfa/a	1.21	(upper 1.34)	SCR50	9.03
p=0.05 JNR=30 D-IN4	Pfa/a	22.97	(upper 23.55)	SCR50	5.59
p=0.05 JNR=30 Clip-OS1	Pfa/a	1.28	(upper 1.38)	SCR50	-2.09
p=0.05 JNR=30 Clip-OS1-cert	Pfa/a	0.11	(upper 0.14)	SCR50	7.13
p=0.05 JNR=30 Bin-OS1	Pfa/a	1.20	(upper 1.52)	SCR50	-0.99
p=0.05 JNR=30 Bin-OS1-cert	Pfa/a	0.05	(upper 0.08)	SCR50	8.17
p=0.05 JNR=30 Exc-D-IN1	Pfa/a	1.37	(upper 1.50)	SCR50	nan

I1 True; I2 True; I3 True (Clip-OS1 clean SCR50 -2.32; cert at p=0.02: -0.85 / -0.89)

Verdicts: L1 NOT met, L2 met, L3 NOT met, L4 NOT met, L5 met, L6 met, A1 met, A2 NOT met, A3 met, T1 met, T2 met, T3 met, I1 met, I2 met, I3 met

77 GHz data with real interference (r12/analyze_r12_jku.py); N: no fast-time mitigation, Z: zeroing, ZAR: zeroing with AR(8) gap reconstruction; strata: number of hit chirps in the 24-chirp segment (0, 1-2, 3-5, ≥ 6):

R12 Part J: 8 (station, rx) units; hit rate (test frames): ref 0.000, A 0.454, C 0.046
zeroed fraction (test frames): ref 0.049, A 0.076, C 0.070

run_ref, mitigation N:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.04	1.04 0.00 0.00 0.00	; SCR50 10.01 [10.01]
OS1_8	1.00	1.00 0.00 0.00 0.00	; SCR50 10.80 [10.80]
D-IN1	1.03	1.03 0.00 0.00 0.00	; SCR50 5.91 [5.91]
Clip-OS1	1.00	1.00 0.00 0.00 0.00	; SCR50 6.84 [6.84]
Bin-OS1	0.67	0.67 0.00 0.00 0.00	; SCR50 7.96 [7.96]
Clip-OS1-cert	1.00	1.00 0.00 0.00 0.00	; SCR50 6.84 [6.84]
Bin-OS1-cert	0.67	0.67 0.00 0.00 0.00	; SCR50 7.96 [7.96]

run_ref, mitigation Z:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.07	1.07 0.00 0.00 0.00	; SCR50 10.33 [10.33]
OS1_8	0.98	0.98 0.00 0.00 0.00	; SCR50 11.11 [11.11]
D-IN1	0.99	0.99 0.00 0.00 0.00	; SCR50 6.24 [6.24]
Clip-OS1	0.97	0.97 0.00 0.00 0.00	; SCR50 7.11 [7.11]
Bin-OS1	0.63	0.63 0.00 0.00 0.00	; SCR50 8.32 [8.32]
Clip-OS1-cert	0.97	0.97 0.00 0.00 0.00	; SCR50 7.11 [7.11]
Bin-OS1-cert	0.63	0.63 0.00 0.00 0.00	; SCR50 8.32 [8.32]

run_ref, mitigation ZAR:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.05	1.05 0.00 0.00 0.00	; SCR50 10.21 [10.21]
OS1_8	1.00	1.00 0.00 0.00 0.00	; SCR50 10.97 [10.97]
D-IN1	1.01	1.01 0.00 0.00 0.00	; SCR50 6.09 [6.09]
Clip-OS1	0.99	0.99 0.00 0.00 0.00	; SCR50 7.00 [7.00]
Bin-OS1	0.66	0.66 0.00 0.00 0.00	; SCR50 8.17 [8.17]
Clip-OS1-cert	0.99	0.99 0.00 0.00 0.00	; SCR50 7.00 [7.00]
Bin-OS1-cert	0.66	0.66 0.00 0.00 0.00	; SCR50 8.17 [8.17]

run A, mitigation N:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.09	1.12 1.22 0.56 1.06	; SCR50 15.29 [11.91]
OS1_8	1.16	1.05 1.21 0.83 1.20	; SCR50 16.01 [12.70]
D-IN1	0.81	0.94 1.00 0.49 0.72	; SCR50 11.14 [7.68]
Clip-OS1	1.01	0.92 0.98 1.11 1.05	; SCR50 11.95 [8.66]
Bin-OS1	0.68	0.60 0.65 0.76 0.71	; SCR50 13.08 [9.83]
Clip-OS1-cert	0.11	0.40 0.03 0.00 0.00	; SCR50 nan [13.74]
Bin-OS1-cert	0.06	0.22 0.00 0.00 0.00	; SCR50 nan [14.93]

run A, mitigation Z:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.04	1.14 1.08 1.25 0.99	; SCR50 11.07 [10.32]
OS1_8	0.98	1.04 1.04 1.32 0.94	; SCR50 11.86 [11.14]
D-IN1	0.96	0.99 1.06 0.83 0.93	; SCR50 6.96 [6.11]
Clip-OS1	0.98	0.97 1.18 0.76 0.95	; SCR50 7.86 [7.05]
Bin-OS1	0.62	0.58 0.75 0.62 0.61	; SCR50 9.05 [8.33]
Clip-OS1-cert	0.11	0.42 0.02 0.00 0.00	; SCR50 nan [12.24]
Bin-OS1-cert	0.06	0.22 0.00 0.00 0.00	; SCR50 nan [13.63]

run A, mitigation ZAR:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.07	1.15 1.13 1.04 1.03	; SCR50 10.89 [10.27]
OS1_8	1.04	1.08 1.12 1.11 1.00	; SCR50 11.63 [11.03]
D-IN1	1.01	1.01 1.10 0.69 1.00	; SCR50 6.74 [6.06]
Clip-OS1	1.01	0.93 1.11 0.83 1.02	; SCR50 7.65 [7.03]
Bin-OS1	0.65	0.58 0.84 0.76 0.64	; SCR50 8.82 [8.26]
Clip-OS1-cert	0.11	0.41 0.02 0.00 0.00	; SCR50 nan [12.18]
Bin-OS1-cert	0.06	0.22 0.01 0.00 0.00	; SCR50 nan [13.52]

run C, mitigation N:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.13	1.08 1.31 1.57 0.90	; SCR50 11.32 [10.50]
OS1_8	1.08	1.01 1.33 1.63 0.93	; SCR50 12.08 [11.28]
D-IN1	1.42	1.19 3.56 1.99 0.89	; SCR50 7.15 [6.34]
Clip-OS1	1.29	1.11 2.38 2.00 1.13	; SCR50 8.11 [7.29]
Bin-OS1	0.90	0.75 1.73 1.39 0.95	; SCR50 9.22 [8.42]
Clip-OS1-cert	0.54	0.68 0.39 0.00 0.00	; SCR50 14.52 [10.40]
Bin-OS1-cert	0.33	0.42 0.17 0.00 0.00	; SCR50 15.78 [11.63]

run C, mitigation Z:	Pfa/alpha	(all by hit stratum ['s0', 's1', 's2', 's3'])	; SCR50 all [hit-free stratum]
IN1	1.04	1.05 0.97 1.04 0.98	; SCR50 10.54 [10.36]
OS1_8	0.98	0.98 0.96 0.99 1.00	; SCR50 11.32 [11.15]
D-IN1	1.00	1.00 0.99 0.94 1.03	; SCR50 6.41 [6.22]
Clip-OS1	1.02	1.03 1.03 0.98 1.02	; SCR50 7.26 [7.08]
Bin-OS1	0.64	0.63 0.63 0.56 0.79	; SCR50 8.48 [8.31]

```

Clip-OS1-cert 0.51 | 0.65 0.26 0.00 0.00 ; SCR50 13.94 [ 10.00]
Bin-OS1-cert 0.27 | 0.36 0.08 0.00 0.00 ; SCR50 15.19 [ 11.33]

run C, mitigation ZAR: Pfa/alpha (all | by hit stratum ['s0', 's1', 's2', 's3']) ; SCR50 all [hit-free stratum]
IN1 1.05 | 1.06 0.99 1.05 1.05 ; SCR50 10.41 [ 10.28]
OS1_8 1.01 | 1.01 1.03 1.03 1.03 ; SCR50 11.18 [ 11.04]
D-IN1 1.04 | 1.04 1.10 0.99 1.04 ; SCR50 6.25 [ 6.10]
Clip-OS1 1.02 | 1.02 1.04 1.04 0.96 ; SCR50 7.15 [ 7.01]
Bin-OS1 0.66 | 0.65 0.64 0.63 0.76 ; SCR50 8.32 [ 8.18]
Clip-OS1-cert 0.50 | 0.65 0.26 0.00 0.00 ; SCR50 13.88 [ 9.93]
Bin-OS1-cert 0.30 | 0.39 0.08 0.00 0.00 ; SCR50 15.08 [ 11.22]

J1 (Clip-OS1-cert Pfa <= 1.2 alpha in run C under N and Z): True [N 0.0054, Z 0.0051]
J2 (D-IN1 under N Pfa >= 5 alpha in runs A and C): False [A 0.0081, C 0.0142]
J3 (ZAR improves IN1 SCR50 over Z by >= 1 dB in run A): False [0.18 dB]

```

18 Second radar: full results

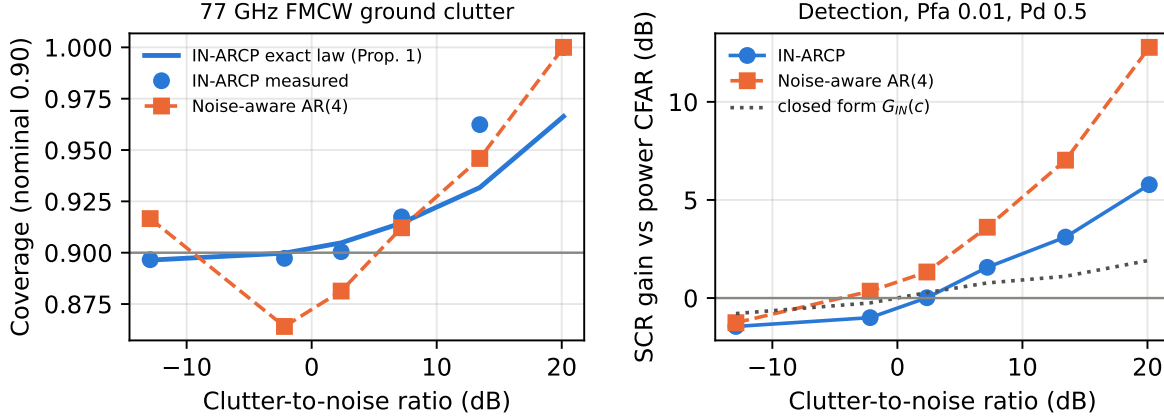


Figure 6: 77 GHz FMCW ground clutter: coverage by CNR class with the exact law, and SCR gain over the local power detector.

```

units 24 (station x Rx x rotation); test episodes per CNR class {'<5': 721095, '-5..0': 265416, '0..5': 229845, '5..10': 68556, '10..20': 31870, '>=20': 338}
mean CNR per class (dB): -12.9 -2.2 2.4 7.2 13.4 20.1

=== 1-alpha = 0.9 ===
J1 marginal coverage: IN1 0.9000 | IN4 0.9000 | NA1 0.8999 | NA4 0.9000 | G1 0.8974 | NA4G 0.8933 | U 0.8998 | RMS 0.9000 | CA16 0.9000 | CAlc 0.9001
J2 coverage by CNR class (mean over units):
IN1 0.897 0.897 0.900 0.917 0.962 1.000 | mean max class deviation 0.066
NA1 0.939 0.850 0.840 0.863 0.882 0.945 | mean max class deviation 0.067
NA4 0.917 0.864 0.881 0.912 0.946 1.000 | mean max class deviation 0.055
U 0.972 0.896 0.824 0.678 0.305 0.017 | mean max class deviation 0.629
CA16 0.893 0.895 0.904 0.942 0.985 1.000 | mean max class deviation 0.086
IN1 exact law 0.896 0.900 0.905 0.914 0.932 0.966 | MAE per unit x class 0.0104

=== 1-alpha = 0.99 ===
J1 marginal coverage: IN1 0.9900 | IN4 0.9900 | NA1 0.9900 | NA4 0.9901 | G1 0.9897 | NA4G 0.9846 | U 0.9899 | RMS 0.9900 | CA16 0.9900 | CAlc 0.9900
J2 coverage by CNR class (mean over units):
IN1 0.989 0.989 0.990 0.995 0.999 1.000 | mean max class deviation 0.009
NA1 0.996 0.983 0.979 0.984 0.989 1.000 | mean max class deviation 0.013
NA4 0.993 0.983 0.987 0.993 0.998 1.000 | mean max class deviation 0.009
U 1.000 0.997 0.991 0.959 0.734 0.245 | mean max class deviation 0.305
CA16 0.989 0.989 0.992 0.998 1.000 1.000 | mean max class deviation 0.010
IN1 exact law 0.989 0.990 0.991 0.993 0.995 0.998 | MAE per unit x class 0.0019

J3 detection (Swirling 1, Pfa 0.01): SCR gain vs CAlc at Pd=0.5 per CNR class, mean [95% CI]; closed form G_IN
G_IN(c) dB -0.79 -0.25 0.28 0.77 1.11 1.92
IN1 -1.45 [-1.79,-1.13] -1.00 [-1.19,-0.83] 0.02 [-0.23, 0.23] 1.57 [ 0.89, 2.20] 3.11 [ 1.97, 4.26] 5.78 [ 5.78, 5.78]
IN4 -1.42 [-1.57,-1.28] -0.85 [-0.99,-0.72] 0.52 [ 0.03, 0.96] 3.19 [ 2.27, 3.98] 6.78 [ 5.08, 8.34] 12.33 [12.33,12.33]
NA1 -2.10 [-2.65,-1.61] 0.65 [ 0.31, 0.93] 1.99 [ 1.06, 2.70] 4.21 [ 2.71, 5.51] 7.81 [ 5.26,10.08] 13.07 [13.07,13.07]
NA4 -1.26 [-1.38,-1.14] 0.35 [ 0.08, 0.64] 1.33 [ 0.78, 1.88] 3.61 [ 2.70, 4.44] 7.03 [ 5.50, 8.59] 12.78 [12.78,12.78]
U -5.48 [-6.02,-5.02] -2.19 [-2.65,-1.80] 0.22 [-0.21, 0.61] 4.11 [ 3.54, 4.65] 11.44 [10.93,11.95] 13.07 [13.07,13.07]
CA16 -0.62 [-0.64,-0.59] -0.49 [-0.55,-0.44] -0.47 [-0.50,-0.46] -0.54 [-0.57,-0.51] -0.61 [-0.67,-0.54] -0.85 [-0.85,-0.85]

J4 interference scenario C (Y-chirp hit rate 0.020; hit episodes 1920, clean 53520)
IN1 coverage 0.90: clean 0.901 hit 0.792 | 0.99: clean 0.990 hit 0.979
NA1 coverage 0.90: clean 0.828 hit 0.584 | 0.99: clean 0.964 hit 0.854
NA4 coverage 0.90: clean 0.870 hit 0.644 | 0.99: clean 0.980 hit 0.926
U coverage 0.90: clean 0.761 hit 0.449 | 0.99: clean 0.950 hit 0.785
CA16 coverage 0.90: clean 0.900 hit 0.719 | 0.99: clean 0.989 hit 0.965

```